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Faculty of Artificial Intelligence

# Bylaws of
# Artificial Intelligence College
# Egyptian Russian University

## Bachelor's degree with credit hour system

## 2022

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## OVERVIEW

Higher education institutions are considered one of the most important and greatest pillars through which the wheels of development and civilized growth are launched, in addition to their important role in instilling the values of society and maintaining the system of values and morals. Performing the role required of it in achieving sustainable development and comprehensive renaissance in various fields and sectors.

Students are the centerpiece of the university's education system; Therefore, the student is one of the main inputs to the educational process, and the graduate is the main output of the educational process. Therefore, the educational process should revolve around the student to achieve its goals and achieve the targeted learning outcomes, and recent trends in the formation of education confirm that the student should not look at It is the recipient of the educational service, but is seen as an essential partner. The design of the programs of this regulation has taken into account the students' acquisition of the skills that must be available in accordance with the national standards for the quality of education that put the student in the center of attention in order to build an Egyptian society that learns, thinks and innovates this vision.

# First: The vision, mission and objectives of the college

## 1. The vision, mission and goals of the college

## Vision

The Faculty of Artificial Intelligence at the Egyptian Russian University is looking forward to excellence in the field of education and scientific research, to achieve a prominent position locally and internationally, and to develop the information society while adhering to professional ethics.

## Mission

The Faculty of Artificial Intelligence at the Egyptian Russian University seeks to prepare specialized and distinguished cadres in the field of computers and artificial intelligence, qualified and equipped with educational, research and applied foundations in the field of specialization in line with quality standards in the fields of computers and artificial intelligence and capable of continuous learning and competition in the labor market within a framework of moral values, and to contribute In developing scientific research, local and international cooperation, and effective participation in serving the community and the surrounding environment.

# Objectives

**1-** Preparing specialists in computers and artificial intelligence as a unique specialization who are qualified with theoretical foundations and application methodologies to qualify them for local, regional and global competition in the development of computer technology and artificial intelligence and its applications.

**2-** Conducting scientific and applied studies in the field of computers and artificial intelligence, foremost of which are those that have a direct impact on achieving sustainable development and keeping pace with the state's trend towards digital transformation.

**3-** Providing scientific and technical advice and assistance to bodies and authorities that use computer technology and artificial intelligence and are interested in industry and decision-making.

**4-** Spreading and deepening awareness in society with the aim of using computer technology and artificial intelligence in the various sectors and institutions of the state, and raising the efficiency of their use.

**5-** Concluding scientific agreements with corresponding bodies and institutions at the local, regional and global levels with the aim of exchanging opinions and conducting research related to the disciplines of computers and artificial intelligence, providing and supporting the means of publishing and scientific research in various fields of specialization.

**6-** Spreading awareness of computer technology and artificial intelligence and deepening it in society with the aim of using computer technology and artificial intelligence in the various sectors and institutions of the state, and raising the efficiency of their use.

**7-** Participation with specialized agencies for the development of software and various smart applications.

**8-** Continuous development in academic programs and educational systems in line with the requirements of preparing a distinguished and competitive graduate.

## Qualifications of Artificial Intelligence Faculty Graduate:

- ✓ Apply knowledge of mathematics, science, and computing concepts in solving computer problems.
- ✓ Identify, formulate and solve computer problems.
- ✓ Exploiting the techniques, skills and modern computing tools necessary for practical computing practices.
- ✓ Designing computing components, processes and systems to meet the required needs within realistic constraints, taking into account the negative impact of computing solutions on society and the environment.
- ✓ Design and conduct experiments, and analyze and interpret data implications.
- ✓ Demonstrate knowledge of contemporary computing issues.
- ✓ Work efficiently within multidisciplinary teams.
- ✓ Demonstrate professional responsibilities and ethical, societal and cultural concerns.
- ✓ Realizing the need for lifelong self-learning.
- ✓ Managing computing projects subject to economic, environmental and social constraints.
- ✓ Meet the requirements of potential employers.

## Second: College departments, specializations, and degrees

## Article (1) College Departments and Academic Degrees:

- **Academic Degrees:**

The Faculty of Artificial Intelligence at the Egyptian Russian University awards a bachelor's degree in one of the following major disciplines:

1. **Department of Artificial Intelligence**
    - Artificial Intelligence Program
2. **Department of Data Science**
    - Data Science Program

## Article (2) Departments Academic areas:

1. **Department of Artificial Intelligence**
   - **Artificial Intelligence includes the following scientific fields:**

     Computer programming and concepts of computer languages and their translators - data structures - analysis and design of algorithms - systems - computer architecture and organization - database systems - operating systems - machine learning and advanced algorithms - software engineering - digital image processing and computer vision systems - artificial intelligence sciences and its applications such as Recommendation systems and expert systems - Artificial intelligence applications in pattern recognition, computer vision, signal analysis, game theory and deep learning - Mathematical, scientific and technical basis for solving problems in the field of artificial intelligence - Development and design of programs and systems in the field of artificial intelligence - Design of computer systems and artificial intelligence systems - Tools used In designing, implementing and evaluating computer and artificial intelligence systems and programs - Department of computing, information and artificial intelligence sciences - Analysis of qualitative and quantitative data - Data handling and analysis in real time - Applied areas related to artificial intelligence - Natural language processing - Data mining and databases - Visual computing - Foundations Image and signal analysis - different applications Knowledge in game theory and natural language processing - principles of software engineering - software systems and software tools - parallel computing - design of knowledge systems and natural language processing systems - machine learning systems - artificial neural networks and deep learning - artificial intelligence algorithms - features, components and patterns of artificial intelligence systems - solution Problems of artificial intelligence systems with commercial or industrial requirements - application of artificial intelligence and computing systems - application of artificial intelligence, computing and information retrieval skills - design, implementation and management of systems and software for artificial intelligence - construction of artificial intelligence software.

## 2. Department of Data Science

- **Data Science includes the following scientific fields:**

  Data analysis, design and presentation - Data science development methodologies - Data science architectures - Data storage and retrieval systems - Database systems - Distributed and parallel database systems - Big data management - Big data decision-making methods - Big data and business management - Intelligent data science systems - Data science center management - Data science development methodologies - Data science applications in various fields - Data science center management - Software and data science quality assurance - Designing information systems that fit with big data - Discovering knowledge within data, especially Big data - Methods and methods for data mining and predicting future changes and how to use them to solve real problems - Applying data science principles to analyze business problems - Collecting structured or unstructured data - Carrying out the process of cleaning, coordinating, processing and organizing data - Building analytical and descriptive models to present Big Data Analytics Services - Tools and techniques for big data analysis.

# Third: The college's registration and study system

## Article (3): Rules of Acceptance and Transfer:

- **College admission requirements**
  - The Faculty of Artificial Intelligence at the Egyptian Russian University accepts students who have obtained a secondary school, the scientific division (mathematics or science), or its equivalent accredited certificates, according to the coordination conditions of the Supreme Council of Universities or the Council of Private and Private Universities.
  - Students of general secondary school (science department) must pass the pre-mathematic course for secondary school science students in mathematics before graduation.
  - Students of secondary school science (mathematics department) must pass the Principles of Biology course if they join the Intelligent Bioinformatics program or the Intelligent Biomedical Informatics program before the specialization.
  - Students can be accepted as listeners on a course - without obtaining a university degree - according to the rules set by the faculties

- **Terms of Transfer to the College**
  - Students may be transferred to the college from university colleges or similar institutes, provided that they obtain the minimum total grades prescribed for admission to the college, in a manner that does not conflict with the rules and regulations set by the Supreme Council of Universities and the Council of Private and National Universities.
  - In the case of acceptance of the student's transfer from the corresponding colleges, a clearing will be made to equalize the courses he studied with the courses offered by the department to which he is transferred, and it is required to be approved by the dean of the college and the university president for it. It is permissible to transfer from one major to another within the college, provided that the conditions for admission to the new major to which he is transferred are met.
  - The CGPA obtained by the student must not be less than 2 out of 4, and the percentage of any of the courses transferred to the student must not be less than 60%.

## Article (4): The language of instruction

- English language, and university requirements courses can be taught in another language, provided that the exam is in the same language as the course is taught

## Article (5) Study System:

a) The study depends on the deliberate hours system, and the credit hour is an academic unit of measurement to determine the course weight.
b) The study year consists of two regular semesters - the fall semester and the spring semester - and an optional summer semester for the student, and it is held according to the capabilities of the college and in return for a fee determined by the college and approved by the university
c) The period of regular semesters is 17 weeks, including the period of holding exams
d) The summer semester is a compressed semester where the number of weekly hours of the course is doubled, its duration is 8 weeks including the period of holding exams
e) The maximum period of study in the college is eight academic years (sixteen regular semesters, fall and spring), noting that suspension semesters that were approved by the College Council and the University Council are not counted among the permitted semesters.
f) Definition of the credit hour: the student obtains a number of credit hours according to the number of contact hours per week, and the contact hour is calculated as follows, where each lecture hour is estimated as one credit hour, and each hour of training or laboratory is half a credit hour.
g) In exceptional cases, in accordance with the rules set by the college council and based on the proposal of the academic advisor, the student may bear 3 credit hours more than the maximum limit in cases of graduation and cases determined by the department council after the approval of the dean.

## Article (6) Number of Hours Required for Graduation:

Obtaining a bachelor's degree in any of the disciplines stipulated in Article (1) of these regulations requires the student to successfully pass 144 credit hours (one hundred and forty-four credit hours) distributed over at least six regular semesters.

### Graduate roles are:

- Graduation at the end of the first semester (February round).
- Graduation at the end of the second semester (July round).
- Graduation at the end of the summer semester (September round).

## Article (7) Academic Advising

a) The Vice Dean for Education and Student Affairs, in coordination with the heads of the scientific departments, determines an academic advisor from its faculty members for each student or group of students to assist and guide him academically during the study period with regard to choosing compulsory and elective courses and their previous requirements that he can register in any semester, whether Regular or summer.
b) The college council determines the minimum number of students to be registered in a course and the conditions under which this course can be opened.

c) The academic advisor is of the opinion that the student is fully responsible for the courses in which he registers based on his desire.

## Article (8) Conditions for Graduation

The academic degree is granted when the student fulfills the requirements for obtaining it, as stipulated in the college regulations

a) The student must successfully pass the number of credit hours stipulated in the regulations with a cumulative grade point average (CGPA) of no less than 2.
b) The student must successfully pass all the courses without credit hours stipulated in the college bylaws.
c) The minimum for graduation (obtaining a bachelor's degree) is three academic years, that is, six regular semesters (fall and spring).
d) Passing what the university stipulates as graduation requirements

## Article (9) The percentages of the course distribution

The percentages mentioned and approved by the standard reference standards for the computing sector have been adhered to in the program structure, which are the following:

| | Subject Area | Tolerance % | % |
|---|---|---|---|
| 1 | Humanities, Ethical and Social Sciences (Univ. Req.) | 8-10 | 8.33 |
| 2 | Mathematics and Basic Sciences | 16-18 | 16.66 |
| 3 | Basic Computing Sciences (institution req.) | 26-28 | 27.08 |
| 4 | Applied Computing Sciences (specialisation) | 28-30 | 29.16 |
| 5 | Training | 3-5 | 4.16 |
| 6 | Projects | 3-5 | 4.16 |
| Subtotal | | 84-96 | 89.55 |
| 7 | Optional (Institution character- identifying subjects) | 4-16 | 10.45 |
| **Total** | | 100 | 100 % |

## Article (10) The number of hours required to transfer between the different levels

- When the student joins the college:
  - The new student is enrolled in the first level.
  - A student transferred from another college is enrolled in the same level as the number of hours he passed in the college from which he was transferred, based on a set-off and as shown in the following table.

- The student can move from one level to another at the beginning of each semester, according to the hours passed, as shown in the following table:

| Academic level | Definition of the student | The number of credit hours the student has successfully passed |
|---|---|---|
| First | Freshman | **34** |
| Second | Sophomore | **73** |
| Third | Junior | **109** |
| Fourth | Senior | **144** |

## Article (11) Number of hours of registration in different semesters

- **<u>For the regular seasons (fall and spring):</u>**

- The minimum number of credit hours for registration is (9), and the minimum may be exceeded if the number of hours remaining for the student's graduation is less than 9
- Maximum registered hours:
  - 18 credit hours for new students (not transferred from corresponding colleges and institutes) in the first semester of their enrollment in the college
  - 21 credit hours for students with a CGPA at the beginning of the semester that is greater than or equal to 3, and also if the student graduated in the same semester
  - 18 credit hours for students with a CGPA at the beginning of the semester of higher than or equal to 2 and less than 3.
  - 15 credit hours for students with a CGPA at the beginning of the semester above or equal to 1 and less than 2
  - 12 credit hours for students with a CGPA of less than 1 . at the beginning of the semester
  - In addition to the above-mentioned hours, students who wish to register for one additional course they have previously obtained with an incomplete grade are allowed.

- **<u>For the summer semester:</u>**

- The summer semester is optional for the student.
- The maximum number of hours registered for students is 9 credit hours

## Article (12) Times of registration, deletion, addition and withdrawal

- Course registration continues until the end of the second week. After completing the registration process and starting the study, the student can delete one or more courses from the courses he previously registered, or add one or more courses to complete the credit hours he is authorized to study in the semester after the approval of the academic advisor and taking into account the dates that Determined by the

University Council for deletions and additions of some headquarters until the end of the third week.

## Article (13) Conditions for registration on a course

- The student is allowed to study and register in a course based on his/her passing of its prerequisites.

## Article (14) Withdrawal

- A student may withdraw from registration in one or more courses after the approval of the academic advisor until the end of the ninth week, taking into account the minimum number of credit hours. And he must repeat the course in full, study and an exam.
- If the student withdraws from one or more courses after the specified period without a compelling excuse accepted by the College Council, a grade of “failed” in the courses from which he withdrew will be calculated. But if he submits at least one month before the exam with a compelling excuse accepted by the college council, a “W” grade is calculated for him.

## Article (15) “incomplete” assessment

- If the student presents a compelling excuse accepted by the College Council for not attending the final exam for any course, an “incomplete” grade “I” in this course will be counted, provided that he has obtained at least 60% of the semester work grades, and that he has not been deprived of entering final exams
- A student who obtains an "incomplete" grade is allowed to take the final exam only, and the student's final grade is calculated on the basis of the grade obtained in the final exam in addition to the grade previously obtained in the semester work. This is provided that the student takes the final examination during the same academic year or the following academic year from the calculation of the “incomplete” course, otherwise the grade will turn into “withdrawn” “W” and the student must repeat the course in full, studying and exam, without calculating the previous grade. Get it in the semester work.
- If the student submits a compelling excuse accepted by the College Council for not attending the final exam for any course and has not met the condition that he obtains at least 60% of the semester work grades, he will be counted with a “withdrawal” grade of “W” in the course and he must repeat the course. Full study and examination.

## Article (16) Suspension of Enrollment (Withdrawal from the Semester)

- The student may stop his enrollment (total withdrawal from the academic semester) in accordance with the regulations set by the college and university, provided that this does not exceed at least one month before the exam

- A student who did not appear for registration during the period of registration, deletion and addition in the regular classes is considered to have withdrawn from the semester.
- The number of regular semesters a student withdraws from may not exceed four consecutive semesters or 6 separate semesters.

# Fourth: Examination and assessment system

## Article (17) Examination System

a. The maximum mark for each course is (100) one hundred marks, and the minimum mark for success in any course is (50) fifty marks.

The grades for each course are distributed as follows:

1. For the theoretical course:
   - 40% is allocated for the end-of-semester exam, provided that the student obtains at least 30% of the final (written) exam score, provided that he succeeds in the course.
   - 20% for the mid-semester exam.
   - 40% for the periodic exams that the subject professor conducts on a regular basis and for the semester work.
2. For a course that contains practical applications, it is calculated as follows:
   - 40% is allocated for the end-of-semester exam, provided that the student obtains at least 30% of the final (written) exam score, provided that he succeeds in the course.
   - 20% for the mid-semester exam.
   - 20% for the periodic exams that the subject professor conducts on a regular basis and for the semester work.
   - 20% for practical (exams, applications and projects).
3. For the graduation project, it is calculated as follows:
   - 60% of the quarterly business.
   - 40% to evaluate the discussion committee.

b. The minimum passing grade for the course is 50% of the final grade for the course.

c. The time for the end-of-semester exam for any course is two hours.

## Article (18) Project Decision

- The student who has passed 70% of the number of credit hours required for graduation is entitled to register for the project course
- The project is registered in two consecutive regular semesters (fall and then spring).
- The minimum success rate for the project course is 50% of the total course marks, provided that the student obtains 30% of the discussion mark.

## Article (19) Table of Grades and Points for Credit Hours Courses

- The college follows the credit hour system, which is based on the fact that the basic unit is the academic course and not the academic year. The evaluation system is based on the assessment in each course with a points system, which is determined according to the following table:

| النقاط | التقدير | النسبة المئوية للدرجة |
|---|---|---|
| 4 | A+ | Greater than %96 |
| 3.7 | A | 92%- Lower than 96% |
| 3.4 | A- | 88%- Lower than 92% |
| 3.2 | B + | 84%- Lower than 88% |
| 3 | B | 80%- Lower than 84% |
| 2.8 | B- | 76%- Lower than 80% |
| 2.6 | C + | 72%- Lower than 76% |
| 2.4 | C | 68%- Lower than 72% |
| 2.2 | C- | 64%- Lower than 68% |
| 2 | D+ | 60%- Lower than 64% |
| 1.5 | D | 55%- Lower than 60% |
| 1 | D- | 50%- Lower than 55% |
| Zero | F | Lower than 50% |
| Zero | *Abs* | *Absence from the final exam without an excuse acceptable to the College Council* |
| *Without points and the number of hours is not calculated within the cumulative average until after completing the course study, whether with success or failure* | *Con* | *Continuing course in the next semester* |
| | *I* | *incomplete course* |
| | W | *Withdrawing from a course* |

## Article (20) How to Calculate the Cumulative GPA

$$Cumulated\ GPA\ (CGPA) = \frac{\sum_{\forall\ courses} Grade\ Points * Number\ of\ Credit\ hours}{\sum_{\forall\ courses} Number\ of\ Credit\ hours}$$

Rounding to the nearest two decimal number

## Article (21) Pass and fail courses (without credit hours).

- The full mark on the final exam
- Table of grades for those courses:

| المعنى | التقدير |
|---|---|
| Listener | AU |
| Pass | P |
| Fail | F |
| Withdrawal | W |
| Absence from the final exam without an excuse acceptable to the college council | Abs |
| Incomplete (if the course has a year's work) | I |

## Article (22) General assessment schedule (for each semester – upon graduation).

| التقدير العام | المعدل التراكمي |
|---|---|
| ضعيف جدا | أقل من 1 |
| ضعيف | 1- أقل من 2 |
| مقبول | 2 - أقل من 2.5 |
| جيد | 2.5 - أقل من 3 |
| جيد جداً | 3 - أقل من 3.5 |
| ممتاز | 3.5 فأكثر |

## Article (23) Warning:

- The student gets an academic probation if his cumulative cumulative average (CGPA) in any regular semester is less than 2 (except for the student's first semester in the college) and is placed under academic observation.
- A warning is given to the student in the event that the percentage of absenteeism in the course reaches 20% through statements announced in the college.

## Article (24) Repeating a course in which the student had previously failed

- If the student fails a course, he must re-study it and take the exam again.
- His cumulative average is calculated on this basis, with the number of course hours calculated once
- All repetitions and the grade (or grade) obtained by the student in each re-take appear in the student's detailed certificate.
- The student pays for repeating the course equivalent to the fee he pays if he registers for the course in the summer semester.

## Article (25) Repeating a course in which the student had previously passed in order to raise his cumulative GPA to avoid dismissal

- In the event that the student obtains a cumulative average (CGPA) at the beginning of the semester is less than 2 (the student under academic observation), he must raise his average.
- If the student mentioned in the previous point (under academic observation) wishes to repeat a course in which he has already passed in order to raise his cumulative GPA to avoid dismissal, he must re-study and take the exam again, and in this case he gets the highest grade of the grades obtained. In all repetitions and no more than 83 (highest grade in B)
- The course must be affiliated with the level at which the student is enrolled or affiliated to a level lower than the level at which the student is enrolled.
- There is no maximum number of these courses, but the student can repeat any number of courses he has already passed in order to raise his cumulative GPA. (CGPA) to 2
- His cumulative average is calculated on this basis, with the number of course hours calculated once.
- All repetitions and the grade (or grade) obtained by the student in each re-take appear in the student's detailed certificate.
- The student pays for repeating the course equivalent to the fee he pays if he registers for the course in the summer semester.

## Article (26) Repeating a course in which the student had previously passed in order to raise his cumulative GPA for improvement

- If the student wishes to repeat a course in which he has already succeeded in order to raise his cumulative GPA, he must re-study and take the exam again, and in this case he shall obtain the highest grade of the grades obtained in all repetitions, provided that it does not exceed 83 (the highest score in B)
- The maximum limit for repeating any of the courses he has previously passed in order to raise his cumulative GPA for improvement is 3 courses.
- The course must be affiliated to the level at which the student is enrolled or affiliated to a lower level, the level to which the student is enrolled in.
- His cumulative average is calculated on this basis, with the number of course hours calculated once.
- All repetitions and the grade (or grade) obtained by the student in each re-take appear in the student's detailed certificate.
- The student pays for repeating the course equivalent to the fee he pays if he registers for the course in the summer semester.

## Article (27) Rules of Attendance:

- A student entering the final exam for any course that has been registered requires achieving an attendance rate of not less than 75% of the lectures and applications specified for him. In the event that the student exceeds the percentage of absence (more than 25%) in any of the courses he has recorded, the College Council has the right to deprive him of entering the final exam after being warned in writing.
- If the student misses the final exam for any course without an acceptable excuse, he will be given a failing grade (FA) in this course. If the absent student presents before the exam with a compelling excuse accepted by the college council, an incomplete grade (I) is calculated for him in this course, provided that the student obtains at least 60% of the semester work grades.
- A student who obtains an incomplete grade has the opportunity to take the final exam for this course within a maximum of one week from the beginning of the next semester.
- The grade is calculated for the student in the course he is examining in the case of an incomplete grade on the basis of the grade obtained in the final exam in addition to the grade previously obtained in the semester work.
- The College Council, after taking the opinion of the concerned department council, and according to the nature of the academic courses, may decide to teach one or more courses in the hybrid education style, so that the study in the course is 60% face-to-face and 40% in the distance education system, or any other percentage, and the presentation of This is for the University's Education and Student Affairs Council to approve and submit it to the University Council for approval.

## Article (28) Honors

- The student is granted honors if he passes the courses he studied at each academic level with a grade of no less than 3 CGPA not less than very good (i.e. an aggregate cumulative average), and provided that the study period does not exceed four years (eight regular semesters), or The student has failed or was deprived of any course during his studies at the college or college from which he was transferred (if any).

## Article (29) Field Training

- A summer training, at least total 90 hours, provided that the student has passed at least 50% of the number of credit hours required for graduation.
- A faculty member is designated as responsible for training with a number of assisting staff members in order to follow up the participants in training annually and put the evaluation of each of them according to the criteria that are determined by the College Council.
- The training course is not counted within the credit hours, but it is a course without credit hours, and therefore it is not counted within the cumulative total, but it is one of the graduation requirements.

## Article (30) Dismissal from college

- A student who obtains an academic probation is separated in four consecutive regular semesters, or six separate regular semesters (with the possibility of some of them being consecutive).
- The student will be expelled from the college if the maximum period of study in the college has been exceeded, after deleting the enrollment suspension classes.
- A student who is subject to dismissal from study for any of the reasons mentioned above, may be given an additional and final opportunity to register in two consecutive regular semesters in addition to a summer semester, in order to fulfill the graduation requirements, provided that he has passed at least 80% of the total number of hours required for graduation. And that after the approval of the College and University Council.

## Article (31) Ranking of students

Students are ranked based on the overall cumulative grade point average (CGPA), and in case of equality, they are ranked according to the total grades.

## Article (32) Transitional Article

- Students of advanced computing at the College of Management, Professional Technology and Computers are transferred to the College of Artificial Intelligence and are graduated in the latter college due to the issuance of Republican Degree No. (583) for the year 2021 to change the name of the College of Management, Professional Technology, and Computers to become the College of Management, Economics and Business Technology, and the bylaw of the college does not contain the department of advanced computing.
- The bylaw of the Advanced Computing department that students were accepted into in the academic year 2020/2021 and 2021/2022 is considered a complementary part to the current bylaw (the bylaw of the College of Artificial Intelligence) and continues to be applied to students until their graduation.
- It is allowed that any of the students of the Advanced Computing department wish to transfer to one of the programs of the current bylaw (the bylaw of the College of Artificial Intelligence). The necessary clearing is done and the student is charged with the courses he did not study if necessary.

## Article (33) Student Grievances

- Any student has the right to submit a complaint against the result of his examination in one or more courses, no later than one week from the date of announcing the results.
- The student is informed in writing of the outcome of the examination of the grievance submitted by him.

# Fifth: University regulations and traditions and the penalty for violating them

## Article (34) University regulations and traditions and the penalty for violating them

- Every breach of university laws, regulations and traditions is considered a disciplinary offense, in particular:
  1. Acts that violate the college system or university facilities.
  2. Incitement to refrain from attending lessons, exams, and other university works.
  3. Every act inconsistent with good conduct inside or outside the university.
  4. Every breach of the exam system and every cheating or attempting to cheat in the exam.
- Every student who commits or attempts to cheat in an exam and is caught in flagrante delicto shall be taken out by the dean or the head of the exam committee and he shall be denied entry to the exam.
- Disciplinary penalties are:
  1. The alert is verbally or in writing.
  2. Warning.
  3. Denial of some student services.
  4. Deprivation from attending the lessons of one of the courses for a period not exceeding one month.
  5. Dismissal from the college for a period not exceeding one month.
  6. Denial of an exam in one or more courses.
  7. Dismissal from the college for a period not exceeding one semester.
  8. Final dismissal from the university.
- The dean of the college has the right to impose the first six penalties after being approved by the university president.
- The Disciplinary Board has the authority to impose all penalties, and the decision to refer the student to the Disciplinary Board is issued by the University President at the request of the Dean.
- The Student Disciplinary Council is formed as follows:
  1. Dean of the College.
  2. The agent of the college.
  3. One of the faculty members is annually chosen by the faculty council.

## Article (35) Additional Rules

Topics that are not mentioned in the articles of this bylaw are presented to the College Council and approved by the University Council.

## Article (36) The rules of the code system for courses

The code of any course consists of a group of letters in the far left representing the code for the major or department, as shown in the following table:

| Department or Discipline | Code |
|---|---|
| University Requirements | HM |
| Mathematics | MTH |
| Physics | PH |
| Electronic and communication Engineering | ELC |
| Computer Sciences | CS |
| Information Systems | IS |
| Software Engineering | SW |
| Artificial Intelligence | AI |
| Data Science | DS |
| Robotics | ROB |

# Sixth: Study plans

# For

# Departments Programs

# Study plans

# For

# Artificial Intelligence Programs

| A: University Requirements (12 credit hours) | | | | | |
|---|---|---|---|---|---|
| 1. Compulsory courses (10 Credit hours) | | | | | |
| **Code** | **Course title(s)** | **Lectures** | **Lab/Tutorial** | **Credits** | **Pre-requisite(s)** |
| HM001 | English language 1 | 2 | - | 2 | |
| HM002 | English language 2 | 2 | - | 2 | HM001 |
| HM003 | Human rights& anticorruption | 2 | - | 2 | |
| HM004 | Russian language 1 | 2 | - | 2 | |
| HM005 | Russian language 2 | 2 | - | 2 | HM004 |
| 2. Elective courses (2 Credit hours) | | | | | |
| HM006 | Scientific thinking | 2 | - | 2 | |
| HM007 | Technical writing | 1 | | 1 | |
| HM008 | Scientific Research Methodology. | 2 | - | 2 | |
| HM009 | History of Egypt | 1 | | 1 | |
| HM010 | Russian Language3 | 2 | - | 2 | HM005 |

| B: Math & Basic Science (21 credit hours) | | | | | |
|---|---|---|---|---|---|
| **Code** | **Course title(s)** | **Lectures** | **Lab/Tutorial** | **Credits** | **Pre-requisite(s)** |
| MTH101 | Mathematics 1 | 2 | -/2 | 3 | |
| MTH102 | Linear Algebra | 2 | -/2 | 3 | |
| MTH103 | Mathematics 2 | 2 | 2/- | 3 | MTH101 |
| MTH104 | Probability and Statistics 1 | 2 | -/2 | 3 | MTH101 |
| PH101 | Physics | 2 | 2/- | 3 | |
| ELC101 | Electronics | 2 | 2/- | 3 | PH101 |
| ELC201 | Logic Design | 2 | 2/- | 3 | ELC101 |
| **Math & Basic Science Elective Courses (3 Credit hours)** | | | | | |
| **Code** | **Course title(s)** | **Lectures** | **Lab/Tutorial** | **Credits** | **Pre-requisite(s)** |
| MTH201 | Mathematics 3 | 2 | 2/- | 3 | MTH103 |
| MTH202 | Probability and Statistics 2 | 2 | -/2 | 3 | MTH104 |
| MTH203 | Numerical Analysis | 2 | -/2 | 3 | MTH101 |

| C: Basic Computer Science (39 credit hours) | | | | | |
|---|---|---|---|---|---|
| **Code** | **Course title(s)** | **Lectures** | **Lab/Tutorial** | **Credits** | **Pre-requisite(s)** |
| CS101 | Introduction to Computers Science | 2 | 2/- | 3 | |
| IS101 | Fundamentals of Information Systems | 2 | 2/- | 3 | |
| CS102 | Structured programming | 2 | 2/- | 3 | CS101 |
| CS103 | Discrete Structure | 2 | 2/- | 3 | |
| IS202 | System Analysis and Design | 2 | 2/- | 3 | IS101 |
| IS201 | Database Systems | 2 | 2/- | 3 | IS101 |
| CS201 | Object Oriented Programming | 2 | 2/- | 3 | CS102 |
| SW201 | Software Engineering | 2 | 2/- | 3 | CS102 |
| CS202 | Signal and System | 2 | -/2 | 3 | MTH103 |
| CS203 | Data Structures | 2 | 2/- | 3 | CS201 |
| CS204 | Operating Systems | 2 | 2/- | 3 | CS102 |
| CS205 | Algorithms | 2 | 2/- | 3 | CS102 |
| AI201 | Introduction to Artificial Intelligence | 2 | 2/- | 3 | MTH102 |

| D: Major Requirements (69 credit hours) | | | | | |
|---|---|---|---|---|---|
| 1. Compulsory Courses (48 Credit hours) | | | | | |
| **Code** | **Course title(s)** | **Lectures** | **Lab/Tutorial** | **Credits** | **Pre-requisite(s)** |
| SW303 | User Interface Design | 2 | 2/- | 3 | --- |
| CS302 | Computer Architecture and Organization | 3 | 2/- | 3 | ELC201 |
| AI301 | Machine Learning | 2 | 2/- | 3 | AI201 |
| DS307 | Cloud computing | 2 | 2/- | 3 | CS204 |
| AI302 | Natural Language Processing | 2 | 2/- | 3 | AI201 |
| AI303 | Speech Recognition | 2 | 2/- | 3 | AI201 |
| AI304 | Computer vision | 2 | 2/- | 3 | AI201 |
| AI305 | Pattern Recognition | 2 | 2/- | 3 | AI201 |
| AI306 | Computational Perception | 2 | 2/- | 3 | AI201 |
| AI401 | Intelligent Algorithms | 2 | 2/- | 3 | AI301 |
| AI402 | Computational Cognitive Systems | 2 | 2/- | 3 | AI306 |
| AI403 | Deep Learning | 2 | 2/- | 3 | AI301 |
| AI405 | Multi Agent Systems | 2 | 2/- | 3 | AI402 |
| AI406 | AI Applications | 2 | 2/- | 3 | AI401/ AI304 |
| AI404 | Graduation Project 1 | 2 | 2/- | 3 | --- |
| AI407 | Graduation Project 2 | 2 | 2/- | 3 | AI404 |
| 2. Elective Courses (21 Credit hours) | | | | | |
| **Code** | **Course title(s)** | **Lectures** | **Lab/Tutorial** | **Credits** | **Pre-requisite(s)** |
| CS301 | Operations Research | 2 | 2/- | 3 | MTH202 |
| ROB302 | Fundamental of Cognitive Interaction with Robots | 2 | 2/- | 3 | AI**201** |

| | | | | | |
|---|---|---|---|---|---|
| AI307 | Computational Learning Theory | 2 | 2/- | 3 | CS203 |
| AI308 | Language Modeling | 2 | 2/- | 3 | MTH201 |
| CS307 | Computer Graphics | 2 | 2/- | 3 | MTH102/ CS102 |
| CS309 | Embedded Systems | 2 | 2/- | 3 | CS204 |
| CS303 | Image Processing | 2 | 2/- | 3 | CS202 |
| SW305 | Software Development for Mobile Devices | 2 | 2/- | 3 | SW201 |
| AI309 | User Models | 2 | 2/- | 3 | AI301 |
| AI310 | Handwriting Recognition | 2 | 2/- | 3 | AI201 |
| AI311 | Expert Systems | 2 | 2/- | 3 | AI201 |
| AI312 | Architecture of Intelligence | 2 | 2/- | 3 | AI201 |
| AI313 | Intelligent Systems | 2 | 2/- | 3 | AI201 |
| AI314 | Artificial Intelligence in Games | 2 | 2/- | 3 | AI201 |
| AI408 | Cognitive Modeling | 2 | 2/- | 3 | AI307 |
| AI409 | Cognitive Engineering | 2 | 2/- | 3 | AI307 |
| AI410 | Cognitive Natural Networks | 2 | 2/- | 3 | AI307 |
| AI411 | Language and Speech Technology | 2 | 2/- | 3 | AI308 |
| AI412 | Statistical language Modeling | 2 | 2/- | 3 | AI302 |
| AI413 | AI for Robotics | 2 | 2/- | 3 | AI401 |
| AI414 | Intelligent Autonomous Robotics | 2 | 2/- | 3 | AI401 |
| SW401 | Software Testing & Quality Assurance | 2 | 2/- | 3 | SW201 |
| CS403 | Advanced Knowledge Representation and Reasoning | 2 | 2/- | 3 | AI301 |
| AI415 | Selected topics in AI (1) | 2 | 2/- | 3 | AI201 |
| AI416 | Selected topics in AI (2) | 2 | 2/- | 3 | AI415 |

# ***Level 1***

| Code | Course title(s) | Credits |
|---|---|---|
| **Semester 1** | | |
| MTH101 | Mathematics 1 | 3 |
| MTH102 | Linear Algebra | 3 |
| PH101 | Physics | 3 |
| CS101 | Introduction to Computers Science | 3 |
| IS101 | Fundamentals of Information Systems | 3 |
| HM001 | English language 1 | 2 |
| **Total Credits** | | **17** |
| **Code** | **Course title(s)** | **Credits** |
| **Semester 2** | | |
| ELC101 | Electronics | 3 |
| MTH103 | Mathematics 2 | 3 |
| MTH104 | Probability and Statistics 1 | 3 |
| CS102 | Structured programming | 3 |
| CS103 | Discrete Structure | 3 |
| HM002 | English language 2 | 2 |
| **Total Credits** | | **17** |

# *Level 2*

| Code | Course title(s) | Credits |
|---|---|---|
| **Semester 3** | | |
| IS201 | Database Systems | 3 |
| IS202 | System Analysis and Design 1 | 3 |
| ELC201 | Logical Design | 3 |
| CS201 | Object oriented Programming | 3 |
| CS202 | Signal and System | 3 |
| HM003 | Human Rights & anticorruption | 2 |
| HM004 | Russian language 1 | 2 |
| **Total Credits** | | **19** |
| **Code** | **Course title(s)** | **Credits** |
| **Semester 4** | | |
| CS203 | Data Structure | 3 |
| CS204 | Operating Systems | 3 |
| AI201 | Introduction to Artificial Intelligence | 3 |
| CS205 | Algorithms | 3 |
| SW201 | Software Engineering | 3 |
| HM005 | Russian language 2 | 2 |
| --- | Math & Basic Science Elective Course | 3 |
| **Total Credits** | | **20** |

# ***Level 3***

| Code | Course title(s) | Credits |
|---|---|---|
| **Semester5** | | |
| CS302 | Computer Architecture and Organization | 3 |
| SW303 | User Interface Design | 3 |
| AI301 | Machine Learning | 3 |
| DS307 | Cloud computing | 3 |
| --- | Major elective 1 | 3 |
| --- | Major y elective 2 | 3 |
| **Total Credits** | | **18** |
| **Code** | **Course title(s)** | **Credits** |
| **Semester6** | | |
| AI302 | Natural Language Processing | 3 |
| AI303 | Speech Recognition | 3 |
| AI304 | Computer vision | 3 |
| AI305 | Pattern Recognition | 3 |
| AI306 | Computational Perception | 3 |
| --- | Major elective 3 | 3 |
| **Total Credits** | | **18** |

# *Level 4*

| Code | Course title(s) | Credits |
|---|---|---|
| **Semester7** | | |
| AI401 | Intelligent Algorithms | 3 |
| AI402 | Computational Cognitive Systems | 3 |
| AI403 | Deep Learning | 3 |
| AI404 | Graduation Project 1 | 3 |
| ----- | Major elective 4 | 3 |
| --- | Major elective 5 | 3 |
| **Total Credits** | | **18** |
| **Code** | **Course title(s)** | **Credits** |
| **Semester 8** | | |
| AI405 | Multi Agent Systems | 3 |
| AI406 | AI Applications | 3 |
| AI407 | Graduation Project 2 | 3 |
| --- | Major elective 6 | 3 |
| --- | Major elective 7 | 3 |
| -- | University elective | 2 |
| **Total Credits** | | **17** |

# Courses Descriptions

| A. University Requirements Courses | | |
|---|---|---|
| HM001 | English language 1 | 2 Cr. Hrs |
| Introduction - characteristics of technical English language – revision of English grammar – some styles of writing – characteristics of effective sentences – common faults in writing of sentences in English language – construction of paragraphs: main idea – methods of presentation of main idea – types of paragraphs. In addition, reading activities and general vocabulary. | | |
| HM002 | English language 2 | 2 Cr. Hrs |
| Exercises related to scientific topics - Development of student's knowledge in language and ability for reading and understanding - Developing of understanding and translation abilities of the student - Developing of listening and speech abilities of the student - exercises on writing technical topics - readings in scientific books - methods of search - technical writing. | | |
| HM003 | Human rights& anticorruption | 2 Cr. Hrs |
| This course contains an overview of anti-corruption. It aims to raise awareness and deal with corruption as a matter of public policy. It also includes an introduction to basic human rights under international and national law. This course includes interactive parts that were produced electronically by the National Center for E-Learning of the Supreme Council of Universities. | | |
| HM004 | Russian language 1 | 2 Cr. Hrs |
| Introduction to phonetics. Idea of stress. System of consonants and vowels. differentiation between sound and letter. Phonetic rules of pronunciation of vowels "o" and "e" while unstressed. Pronunciation of voiced consonants as voiceless at the end of the word. | | |
| HM005 | Russian language 2 | 2 Cr. Hrs |
| This courses covers Development of conversational skills on such topics as<br>"meeting people", "introduction of oneself", "family". Grammatical constructions with such expressions as "How many?", "What time is it now?", "How old are you?", "How much?" Present tense. Genitive case. Personal pronouns. | | |
| **University Elective courses** | | |
| HM006 | Scientific thinking | 2 Cr. Hrs |
| This course is required for all students. It emphasizes the unifying aspects of the scientific approach to the study of nature and human behavior. About one-third of the course is devoted to a discussion of the nature of scientific inquiry and investigation. The course focuses on the process of fact identification and concept formation and testing. In the | | |

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| remainder of the course students are exposed to applications of the approach in various disciplines. The course sets some of the major concepts and theories of science into a broad historical, philosophical, and cultural context and traces the development of these theories and concepts to their present status. This serves the double purpose of acquainting the students with the appropriate setting in which a given idea gains relevance and exposing them to the evolution of current methods of investigation. | | |
| HM007 | Technical writing | 1 Cr. Hrs |
| This course contains an introduction to writing numbers, signs, abbreviations and equations. The course includes teaching the student the basics of writing paragraphs (Paragraph), how to start writing, writing the summary and summary, and organizing writing for the topic, as well as reviewing and editing texts. This course provides the student with instruction in various forms of writing, such as correspondence in the field of specialization, proposals, reports, and examples References and table. As well as how to make a final writing review | | |
| HM008 | Scientific Research Methodology. | 2 Cr. Hrs |
| The course is designed to provide students a solid foundation for designing and conducting their own research and to enable them to become knowledgeable individuals of prior academic research results. The course provides an introduction to scientific research methodology and techniques. The course contains the main issues related to research design; quantitative and qualitative research approaches and related methods; selected number of research techniques in social sciences. The course also provides training to students about how to prepare a research proposal prior to conducting their graduation project (dissertation). | | |
| HM009 | History of Egypt | 1 Cr. Hrs |
| This course examines the history of ancient Egypt from the rise of the Egyptian state to the end of the 18th Dynasty in the New Kingdom, a period of more than 1500 years. This span encompasses three separate “golden ages” of Egyptian history and culture, from the pyramid builders of the Old Kingdom through the Middle Kingdom renaissance to the expansionist era of the early New Kingdom. By looking at both political and social history this course will allow students to examine the differences as well as the similarities between these periods and come to a better understanding of the factors, internal and external, which influenced Egyptian historical development. | | |
| HM010 | Russian Language3 | 2 Cr. Hrs |
| Development of listening, conversational, reading and writing skills. System of tenses in Russian grammar. Accusative case constructions. Instrumental case constructions. Personal and possessive pronouns. Main purpose – development of conversational skills on the basis of grammar material. Such topics as "university", "shop", "library", "hospital". Information about countries. Such topics as "Moscow", "Cairo" | | |

| Math & Basic Science Courses | | |
|---|---|---|
| MTH101 | Mathematics 1 | 3 Cr. Hrs |
| The course contains, Limits and continuity, Differentiation, Taylor's series, parametric curves, Partial Differentiation, trigonometric functions, Applications of differentiation, Integration, Techniques of integration, improper integral, applications of integration, and vectors in the plane. | | |
| MTH102 | Linear Algebra | 3 Cr. Hrs |
| This course concentrates on the mathematical theory and methods of linear algebra. This course contains: Indeterminate forms, Infinite series, vectors, curves and surfaces in space, Binomial theorem, matrices and operations, homogenizes and non-homogenous linear equations, Determinants, vector spaces and subspaces, eigenvalues and eigenvectors, matrix diagonalization, partitioned matrices, quadratic forms, and applications of linear algebra. | | |
| MTH103 | Mathematics 2 | 3 Cr. Hrs |
| Basic definition of differential equations- First order differential equations –second and higher order differential equations- Laplace Transformation- Solving differential equations using Laplace Transform- Solving differential equations using series-systems of differential equations- Fourier series- Partial differential equations- Maxima and Minima of functions of two variables- Double Integral and Triple Integral and its applications. | | |
| MTH104 | Probability and Statistics 1 | 3 Cr. Hrs |
| The course contains descriptive statistics, Visual representations of data, Probability theory and statistical inference, random variables, and probability distributions. Estimation and confidence intervals, Correlation, and regression. | | |
| PH101 | Physics | 3 Cr. Hrs |
| The course contains the electric field, Gauss's law, electric potential, capacitance, DC circuits, magnetic fields, Faraday's and Ampere's laws, time-varying fields, Maxwell's equations in integral form, and alternating currents, which earns the student cognitive and technical skills in physics. | | |
| ELC101 | Electronics | 3 Cr. Hrs |
| Introduction, Insulators, Conductors, Semiconductors, Resistors, Capacitors, Coils, Diodes, Application of diodes, Rectifiers, Power supplies, LEDs, Voltage regulators, Transistors, Amplifiers, Timers, Applications. | | |
| ELC201 | Logic Design | 3 Cr. Hrs |

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| This course contains all topics related to logical design, such as a Review on number systems, Coded number systems, logical functions, and their minimization, NAND and NOR gates, Mutual circuit design, adders, subtractors, decoders, encoders, multiplexer/demultiplexer, memory elements: latch and SR-FF, JK-FF, D-FF, T-FF, counters, programmable logic design (PLD) and (FPGA). | | |
| **Math & Basic Science Elective Courses** | | |
| MTH201 | Mathematics 3 | 3 Cr. Hrs |
| This course contains basic definitions of the scalar product of two vectors- the cross product of two vectors- Triple scalar (mixed) product of three scalars- scalar functions and vector functions- vector field- derivative differential operator- the divergence of a vector- curl of a vector- line integrals - surface integral- Gauss divergence theorem - Stokes theorem- Complex functions- Analytic functions- Complex integral. | | |
| MTH202 | Probability and Statistics 2 | 3 Cr. Hrs |
| Types of statistics, Population versus sample, basic terms - numerical descriptive measures for populations and samples, for ungrouped and grouped data: Measures of Central Tendency, Measures of Dispersion, And Measures of Position –Some special discrete distribution( Binomial-Geometric-Negative-Hyper geometric-Poisson, etc.) and continuous distributions (Uniform-Exponential-Weibull-Gamma-Beta-Normal-log normal-Cauchy) , sampling distribution for the sample means – sample proportions and sample variations-center limit theorem – hypothesis testing. | | |
| MTH203 | Numerical Analysis | 3 Cr. Hrs |
| This course contains Numerical Computing and Computers - Solving of Linear systems- Solving Non-linear Equations-Interpolation and curve fitting- approximation of Functions-Finite Difference Numerical Differentiation and Numerical Integration- Numerical solution of ordinary differential equations. | | |

| Basic Computer Science Courses | | |
|---|---|---|
| CS101 | Introduction to Computers Science | 3 Cr. Hrs |
| This course is designed to introduce various types of computers, Computer hardware and software components, different between hardware and software, numbering systems which include binary, decimal, octal, and hexadecimal system. In addition, the course will cover introduction to computer networks, and the main components and protocols used to access the internet. | | |
| IS101 | Fundamentals of Information Systems | 3 Cr. Hrs |
| This course includes the following topics: the components of information systems, the different types of information systems, the Information Technology (IT) and its impact on information systems, the systems development life cycle (SDLC), and how IS effect on competitive advantages in business. | | |
| CS102 | Structured programming | 3 Cr. Hrs |
| Fundamental programming constructs: Syntax and semantics of a higher-level language; variables, types, expressions, and assignment − Simple I/O − Conditional and iterative control structures − Functions and parameter passing − Structured decomposition − Algorithms and problem-solving: Problem-solving strategies; the role of algorithms in the problem-solving process; implementation strategies for algorithms; debugging strategies. | | |
| CS103 | Discrete Structure | 3 Cr. Hr |
| Introduction to logic and proofs − Fundamental structures: Functions; relations; sets; cardinality and countability − Boolean algebra − Propositional logic: Logical connectives; truth tables; normal forms; validity − Elementary number theory: Factorability; properties of primes; greatest common divisors and least common multiples; Euclid's algorithm; modular arithmetic; the Chinese Remainder Theorem − Basics of counting: Counting arguments; pigeonhole principle; permutations and combinations; binomial coefficients − Predicate logic: Universal and existential quantification; modus ponens and modus tollens; limitations of predicate logic − Recurrence relations: Basic formulae; elementary solution techniques − Graphs and trees: Fundamental definitions; simple algorithms; traversal strategies; proof techniques; spanning trees; applications. | | |
| IS202 | System Analysis and Design | 3 Cr. Hrs |
| The aim of this course is to cover the topics related to the upper phases of the Information Systems Development Life Cycle, which are the following: Planning of IS, the detailed analysis of IS, and the Conceptual design of IS. Methodologies, techniques, activities, tasks, deliverables and practical experiences related to the execution of these phases will be covered in the course. | | |
| IS201 | Database Systems | 3 Cr. Hrs |
| This course is the main course in information system. In this course students learn the concepts of database and the difference between file database and relational database. This course also covers the following topics: Phases of database design, Conceptual database design, Classification, specialization, and aggregation abstraction, The Entity-Relationship model, Extended Entity-Relationship model, View design | | |

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| in conceptual schema, Conceptual schema integration, Transforming conceptual schema to relations, Logical database design, Characteristics of good relation schema, Anomalies in relational schema, Function dependencies, and Normalization. The student will learn SQL basic and the advanced concepts of Structured Query Language (SQL). | | |
| CS201 | Object oriented Programming | 3 Cr. Hrs |
| This course contains object-oriented programming principles and techniques. Topics include object-oriented programming concepts, such as classes, objects, methods, interfaces, packages, inheritance, encapsulation, polymorphism, and information hiding. The student will apply techniques using a modern object oriented implementation language. | | |
| SW201 | Software Engineering | 3 Cr. Hrs |
| This course provides an overview of software engineering. This course contains the following topics: Software development life cycle, Project teams, documentation, and group dynamics. Software cost estimation, Requirements engineering and design, Data modeling, object modeling, and object-oriented analysis, Object-oriented programming and design, Software testing, quality assurance, and Software maintenance. | | |
| CS202 | Signal and System | 3 Cr. Hrs |
| Digital Signal Processing (DSP) is concerned with representing signals in digital form and processing these signals and the information they carry. Topics includes: Introduction to continuous time and discrete time signals and systems, linear time invariant systems, Fourier series representation for continuous and discrete time signals, Fourier transform for continuous and discrete time signals, Sampling theorem, Laplace transform, Z-Transform, transfer function, Digital signal convolution, Digital filter design: Finite impulse response, Infinite impulse response. Adaptive digital filters: Concepts, Algorithms, Applications. | | |
| CS203 | Data Structures | 3 Cr. Hrs |
| This course contains topics that include: built-in data structures. Stacks, queues, linked lists, and tree structures. Sorting algorithms, searching algorithms, and hashing. Abstract data types, this course helps students to analyze, evaluate and deal with data. | | |
| CS204 | Operating Systems | 3 Cr. Hrs |
| This course examines types of operating systems. Operating Systems structures: system components and services, also covers the classical internal algorithms and structures of operating systems, including CPU scheduling, memory management, and device management. Considers the unifying concept of the operating system as a collection of cooperating sequential processes. Covers topics including file systems, virtual memory, disk request scheduling, concurrent processes, deadlocks, security, and integrity | | |
| CS205 | Algorithms | 3 Cr. Hrs |

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| This course is concerned with how to think to solve the problems and how to analysis and design algorithms. In this course the students learn different techniques for designing algorithms, advanced sorting and searching methods, Graph algorithms, Notion of an algorithm, Space and time complexities of an algorithm. In addition, they will aware with Fundamental design paradigms: Divide and conquer, Branch and bound, Backtracking dynamic programming greedy methods and Simulation. binary search trees; representations of graphs; depth- and breadth-first traversals − Recursion and divide-and-conquer strategies – Basic algorithmic strategies: Brute-force algorithms; greedy algorithms. | | |
| AI201 | Introduction to Artificial Intelligence | 3 Cr. Hrs |
| The aim of the module is to introduce students to the principal theory and techniques of Artificial Intelligence – AI and build an awareness of its applications and current research trends. The module also aims to develop a broad appreciation of the nature of the problems associated with AI and introduce common techniques that are used to address them. | | |

| Major Requirements Courses | | |
|---|---|---|
| SW303 | User Interface Design | 3 Cr. Hrs |
| Psychological principles of user interface design. Evaluation of user interfaces. Usability engineering. Task analysis, user-centered design, and prototyping. Conceptual models and metaphors. Software design rationale. Design of windows, menus, and commands. Voice and natural language I/O. Response time and feedback. Color, icons, and sound. Internationalization and localization. User interface architectures and APIs. Case studies and project. | | |
| CS302 | Computer Architecture and Organization | 3 Cr. Hrs |
| In Basic computer organization and design: Computer instructions and their codes, timing and control, execution of instructions. Input, output and interrupt. Assembly language: Programming loops, programming arithmetic & logic operations, subroutines, I/O programming. Central processor organization: Processor bus organization. Arithmetic logic unit, stack organization. Instruction formats. Addressing modes. Data transfer and manipulation, program control. Micro-program control organization: Control memory. Address sequencing. Arithmetic processor design and algorithms: Comparison and subtraction of unsigned binary numbers, addition and subtraction algorithms, multiplication and division algorithms. Input/output organization: Peripheral devices, asynchronous data transfer, direct memory access. Memory organization: Auxiliary memory, virtual memory, cache memory, memory management hardware. Pipeline, vector processing, and Multiprocessors. | | |
| AI301 | Machine Learning | 3 Cr. Hrs |
| This course provides a broad introduction to machine learning and statistical pattern recognition. Topics include: supervised learning (generative/discriminative learning, parametric/non-parametric learning, neural networks, support vector machines); unsupervised learning (clustering, dimensionality reduction, kernel methods); learning theory (bias/variance tradeoffs, practical advice); reinforcement learning and adaptive control. The course will also discuss recent applications of machine learning | | |
| DS307 | Cloud computing | 3 Cr. Hrs |
| The transition from a Classic Data Center environment to Virtual Data Center in your journey to the Cloud ، virtualization at each layer – compute ،storage ،network ،desktop ،and application – along with business continuity in a VDC environment. Cloud computing basics ،infrastructure components ،service management activities ،security concerns ،and considerations for Cloud adoption. | | |
| AI302 | Natural Language Processing | 3 Cr. Hrs |
| This course contains an overview of NLP with a major focus on probability and machine learning techniques. The machine learning-based methodology applied to language data with assessment of these experiences is essential. If the course includes an overview of typical NLP applications, such as information | | |

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| extraction, machine translation, question-answer systems, and more focus on one of these applications. In addition, a typical NLP system, such as marking, analysis, and identification of the named entity, also extracts the relationship. | | |
| AI305 | Pattern Recognition | 3 Cr. Hrs |
| The main goal of this course is to teach students the foundations of pattern recognition, learn about how patterns are recognized on the computer, pattern recognition broad applications in various aspects of our day-to-day life. The course aims to provide you with sufficient background in the theoretical concepts behind stages of pattern recognition systems, and understand and implement the algorithms used in various phases of pattern recognition systems, including data acquisition, pre-processing, segmentation, feature extraction and classification. Objectives include enabling students to understand the techniques used recognize patterns, such as statistical approaches, data clustering, neural networks. | | |
| AI306 | Computational Perception | 3 Cr. Hrs |
| The course contains fundamental facts and concepts of perceptual psychology and sensory neuroscience. This is an interdisciplinary field of science, crossing the boundaries between psychology, neuroscience, and engineering. The main goal of the course is to introduce computational perception on both theoretical and practical levels. Participation in small groups to design, implement, and evaluate a prototype of a human-computer interaction system that uses one or more of the techniques covered in this course. | | |
| AI303 | Speech Recognition | 3 Cr. Hrs |
| This course contains an introduction to speech production; general properties of speech signal; Time domain processing of speech; Frequency domain processing of speech; Linear prediction analysis; Cepstral analysis; Feature extraction for speech processing; Introduction to statistical speech recognition; Introduction to speech coding. | | |
| AI304 | Computer vision | 3 Cr. Hrs |
| This course contains the basic concepts in computer vision. It will introduce students to a number of realistic applications that are important to our daily lives. It also includes themes, finding known models in images, recovering the depth of the stereo, camera calibration, image stabilization, mechanical alignment, tracking, boundary detection, as well as recognition. | | |
| AI401 | Intelligent Algorithms | 3 Cr. Hrs |
| This course contains topics that help improve artificial intelligent algorithms such as artificial neural networks, support vector machine, and application to solve them. This course examines a number of concepts and techniques commonly referred to as intelligent algorithms, particularly the discussion of the basic theory of these methodologies when appropriate, which takes an engineering perspective and an approach to intelligent system design, analysis, evaluation, and implementation. Fuzzy systems, genetic algorithms, particle improvement methods, ant colony, and neural networks are prepared. | | |
| AI402 | Computational Cognitive Systems | 3 Cr. Hrs |

This course contains an introduction to the study of the human mind and intelligent systems using an integrative approach. So he explores the nature of intelligence through simulation, robotics, human experiments, and philosophical research. Topics include cognition, awareness, mental representation and models of cognition and mental anatomy (brain analysis) where both are related to cognition.

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| AI403 | Deep Learning | 3 Cr. Hrs |

In this course students are introduced to the architecture of deep neural networks, algorithms that are developed to extract high-level feature representations of data. In addition to theoretical foundations of neural networks, including backpropagation and stochastic gradient descent, students get hands-on experience building deep neural network models with Python. Topics covered in the course include image classification, time series forecasting, text vectorization (tf-idf and word2vec), natural language translation, speech recognition, and deep reinforcement learning. Students learn how to use application program interfaces (APIs), such as TensorFlow and Keras, for building a variety of deep neural networks: convolutional neural network (CNN), recurrent neural network (RNN), self-organizing maps (SOM), generative adversarial network (GANs), and long short-term memory (LSTM).

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| AI405 | Multi Agent Systems | 3 Cr. Hrs |

This course provides the basic theoretical knowledge about intelligent agents and multi-agent systems. The first part of the course covers the different types of agents, their properties and architectures. The second part includes a thorough description of several coordination methods in multi-agent systems. The course also includes a practical component on the lab, in which students have to work in teams to develop a multi-agent system.

- Understand the origins and foundations of distributed computing on the Internet.
- Knowing the possible applications of artificial intelligence for distributed systems on the Internet.
- Understanding the basics of Agent Orientation.
- To analyze a problem distributed in nature to identify the different actors and their functionalities.
- Designing distributed systems using an agent-oriented methodology.
- Extract and represent knowledge about the context necessary to build a distributed application on the Internet that is flexible and robust.
- Designing context ontologies by applying a methodology properly.

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| AI406 | AI Applications | 3 Cr. Hrs |

In this course, Students become familiar with Artificial intelligence image analytics of neurological disorders with MRI and consider the use of machine learning for finding clinical data patterns for forecasting models, Intelligent systems and their applications, AI in Autonomous Vehicles and Drones and learn how self-flying drones are trained using VR, AI and Ocean and AI in sustainable fisheries by explore fisheries management, and learn how AI can keep ocean fisheries sustainable

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| AI404 | Graduation Project 1 | 3 Cr. Hrs |

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| Students spend the first semester for searching idea of project and Examination of the possibility of implementing the project idea. Topics are related to either software development/modification or analyzing with the aim of artificial intelligence finding, interesting, and insights. | | |
| AI407 | Graduation Project 2 | 3 Cr. Hrs |
| Students spend the last semester conducting experiments and writing their thesis. Topics are related to either software development/modification or analyzing with the aim of artificial intelligence finding, interesting, and insights. | | |
| **Major Elective Courses** | | |
| CS301 | Operations Research | 3 Cr. Hrs |
| This course helps students for solving problems in different environments that needs decisions. The course cover topics that include: linear programming, Transportation, Assignment, Critical Path Method (CPM) and Project Evaluation and Review technique (PERT). In applications computer packages will be used for business managers and decision making. | | |
| ROB302 | Fundamental of Cognitive Interaction with Robots | 3 Cr. Hrs |
| This course is an emerging field that combines cognitive science, computer science, neuroscience, and robotics. These robots provide a tool for studying cognitive function by embedding brain models on robotic platforms. Because embodied models capture the complete system (i.e. the interaction between brain, body, and environment), cognitive robotic experiments can increase our understanding of how the brain gives rise to complex behavior. In this course, students learned concepts of embodiment, robot construction, and computer programming. In the lecture portion of the course, we discussed case studies of cognitive robotics. In the lab portion of the course, students constructed robots, using the LEGO Mindstorms NXT kit, and program these robots to perform different behaviors. | | |
| AI307 | Computational Learning Theory | 3 Cr. Hrs |
| This course contains an overview of the types of problems, and techniques that are usually studied in computational learning theory, and provides a basic arsenal of powerful mathematical tools for analyzing machine learning problems. Topics include: Basics of correct probability approximation (PAC) learning Model, learning to Query, Occam's razor, Online Learning, Promotion, (SVM) Support for machine vectors, statistical queries, in addition to the complexity of Rademacher. | | |
| AI308 | Language Modeling | 3 Cr. Hrs |
| This course contains advanced language models mainly for the problem of speech recognition. This course describes the 'language modeling' task as an example of a statistical analysis of natural language. Demonstrates how Language Modeling helps achieve practical tasks (word prediction, spelling correction, ASR, machine translation). In order to introduce to students, the basic probabilistic tools such as N-grams required to produce and evaluate language models. | | |

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| CS307 | Computer Graphics | 3 Cr. Hrs |
| This course introduces techniques for 2D and 3D computer graphics, including simple color models, homogeneous coordinates, affine transformations (scaling, rotation, translation), viewing transformation, clipping, illumination and shading, texture maps, rendering, high level shader language, video display devices, PHical and logical input devices, hierarchy of graphics software, hidden surface removal methods, Z-buffer and frame buffer, color channels, and using a graphics API. | | |
| CS309 | Embedded Systems | 3 Cr. Hrs |
| Nature of embedded systems, particular problems, special issues; role in information technology; embedded microcontrollers, embedded software; real time systems, problems of timing and scheduling; testing and performance issues, reliability; low power computing, energy sources, leakage; design methodologies, software tool support for development of such systems; problems of maintenance and upgrade. | | |
| CS303 | Image Processing | 3 Cr. Hrs |
| In this course students will learn digital image processing techniques including representation, sampling and quantization, image acquisition, imaging geometry, image transforms, image enhancement, image smoothing and sharpening, and image restoration. More advanced topics include degradation models, image filtering, color image processing, and image segmentation. | | |
| SW305 | Software Development for Mobile Devices | 3 Cr. Hrs |
| This course introduces students to programming technologies, design and development related to mobile applications. Topics include accessing device capabilities, industry standards, operating systems, and programming for mobile applications using an OS Software Development Kit (SDK). Upon completion, students should be able to create basic applications for mobile devices. | | |
| AI309 | User Models | 3 Cr. Hrs |
| This course contains how to design a user model in a structured way, how to select the appropriate tools for designing or innovating a business model, how to present your business model in a single sheet, and how to reason from your needs (“I want to’s”) to a business model solution. | | |
| AI310 | Handwriting Recognition | 3 Cr. Hrs |
| In this course students learn how an automatic handwriting recognizer works. Topics include Handwriting Recognition: Overview, Challenges and Future Trends, Thresholding, Historical Document Processing, how to Design Deep Neural Networks for Handwriting Recognition, and Mathematical Expression Recognition. | | |
| AI311 | Expert Systems | 3 Cr. Hrs |
| This course is an introduction to expert systems, in this course the students learn the methodology used to transfer the knowledge of a human expert into an intelligent program that can be used to solve problems or | | |

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| give advice. Topics include: Introduction to Expert Systems, Representation of Knowledge, Methods of Inference, Reasoning under Uncertainty, Inexact Reasoning, and Design of Expert Systems. | | |
| AI312 | Architecture of Intelligence | 3 Cr. Hrs |
| This course refers to intelligent built environments: dynamic systems with the capacity to respond intelligently and immediately to various human, and environmental stimuli for our benefit. The aim of this course is to explore the world of intelligent built environments and the mutual impact, relationships and evolution they have in the culture and daily lives of people. First-hand exploration involving experiential learning and direct activation of the senses, will be a key component of the course. | | |
| AI313 | Intelligent Systems | 3 Cr. Hrs |
| Application Areas of Intelligent Systems, Intelligent System Architecture, Knowledge Engineering and Control, Languages Used in Expert Systems, Bayesian Interference, Fuzzy Logic, Decision Support Systems, Software tools for developing expert systems and intelligent systems. Robotics: Overview; configuration space; planning; sensing; robot programming; navigation and control. | | |
| BNF305 | Genetic Algorithms | 3 Cr. Hrs |
| The course introduces the concept of genetic algorithms, shows how and why these algorithms work, Evolutionary programming, basic genetic algorithms, populations, fitness evaluation, objective functions, cross-over, mutation, strategies for replacement, schema theory, game playing: prisoners dilemma, gray code, floating point representation, integer representation, non-uniform mutation. The course proceeds to include other adaptive methods, such as simulated annealing, tabu search, and neural networks, especially as they relate to genetic algorithms. Application for genetic algorithms, introduction to fuzzy logic, evaluation of fuzzy rules with genetic algorithms, training of neural networks with genetic algorithms | | |
| AI314 | Artificial Intelligence in Games | 3 Cr. Hrs |
| This course will introduce students to the specific techniques used to simulate artificial intelligence in computer and video games. From the catalog: Techniques to stimulate intelligence in video games: movement, pathfinding with A* search, decision/ behavior trees, state machines, machine learning, tactics. Extend games with your own AI implementations; experience "shootout" contests for the best AI algorithm/implementation. | | |
| AI408 | Cognitive Modeling | 3 Cr. Hrs |
| This course explores the principles of cognition and intelligence in human beings and machines, focusing on how to build computational models that, in essence, think and act like people. The course reviews existing frameworks for such models, studies model development within one particular framework, and discusses how models can be employed in real-world domains. | | |

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| AI409 | Cognitive Engineering | 3 Cr. Hrs |
| This course contains fundamentals of cognitive engineering and user-centered design. The purpose of the course is to explore theoretical and methodological issues in the field of cognitive engineering. Additionally, this course contains: Human-centered design of cognitive tools and work systems, Human-computer interaction, decision making, human error, computer-supported distributed work, and design of decision support. | | |
| AI410 | Cognitive Natural Networks | 3 Cr. Hrs |
| In this course, students learn what is meant by cognitive neural networks and how to explain the mathematical equations that underlie them. students also build neural networks using state of the art simulation technology and apply these networks to the solution of problems. In addition, the course discusses examples of computation applied to neurobiology and cognitive psychology. | | |
| AI411 | Language and Speech Technology | 3 Cr. Hrs |
| This course contains an introduction to the basics of speech technology: especially encoding in speech synthesis, and decoding in speech recognition of ambiguous structures during on-line human-machine interactions. The course will include the possible role of prosody in successful human-machine interactive systems, by investigating how speakers use positive features to encode ambiguous constructions, and how, in turn, listeners decode such usages of prosody. | | |
| AI412 | Statistical language Modeling | 3 Cr. Hrs |
| This course contains Statistical language models (SLMs) assign probabilities to sequences of words, and are used in systems that perform speech recognition, machine translation, and many other tasks. This course will cover both the fundamental technologies that comprise statistical language models, from more classical n-gram models to recent memory-based deep neural networks, along with applications of the techniques to important tasks in artificial intelligence. | | |
| AI413 | AI for Robotics | 3 Cr. Hrs |
| This course presents different methods in Artificial Intelligence, including probabilistic inference, planning and search, localization, tracking, mapping and control, all with a focus on robotics and drones. Extensive programming examples and assignments will apply these methods in the context of building self-driving cars and autonomous vehicles. Apply machine learning techniques such as computer vision and natural language processing for robotics. using the LEGO Mindstorms NXT kit, and program these robots to perform different behaviors.<br>• Industry use cases and case studies in AI for robotics<br>• Robotics architecture<br>• Data acquisition and sensor fusion<br>• SLAM<br>• Computer vision<br>• Deep learning applications for robotics, drones, self-driving cars and autonomous vehicles | | |

- Crack detection robots
- Mobile picking robots
- Collaborative robots

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| AI414 | Intelligent Autonomous Robotics | 3 Cr. Hrs |

This course explored the fundamental problems involved in producing real world intelligent behavior in robots, covering the different information processing methods and control architectures that have been developed and are currently in use, including probabilistic methods and approaches inspired by biological systems. The course is structured around a practical task to develop navigation algorithms on a real robot platform.

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| SW401 | Software Testing & Quality Assurance | 3 Cr. Hrs |

Software Quality Assurance is the organization that takes responsibility for the final verification that the software delivered by a solution team has met all the requirements for use in the business, and achieves the expected level of quality. This course explores testing techniques for obtaining that assurance in all stages of software testing, focusing on the creation and maintenance of test cases, procedures, and scenarios. The course also delineates the testing management responsibilities and roles.

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|---|---|---|
| CS403 | Advanced Knowledge Representation and Reasoning | 3 Cr. Hrs |

Knowledge representation issues, Description logics, Ontology, Non-monotonic reasoning (e.g., non-classical logics, default reasoning), Argumentation, Reasoning about action and change, Temporal and spatial reasoning, Rule-based Expert Systems, Semantic networks, Model-based and Case-based reasoning, Planning: Partial and totally ordered planning, Plan graphs, Hierarchical planning, Planning and execution including conditional planning and continuous planning, Mobile agent/Multi-agent planning.

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| AI415 | Selected topics in AI (1) | 3 Cr. Hrs |

Topics which are not included in the curriculum and seems to be needed should be suggested as an elective course by AI department

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| AI416 | Selected topics in AI (2) | 3 Cr. Hrs |

Topics which are not included in the curriculum and seems to be needed should be suggested as an elective course by AI department

# Study plans

# For

# Data Science Programs

| A: University Requirements (12 credit hours) | | | | | |
|---|---|---|---|---|---|
| 1. Compulsory courses (10 Credit hours) | | | | | |
| **Code** | **Course title(s)** | **Lectures** | **Lab/Tutorial** | **Credits** | **Pre-requisite(s)** |
| HM001 | English language 1 | 2 | - | 2 | |
| HM002 | English language 2 | 2 | - | 2 | HM001 |
| HM004 | Russian language 1 | 2 | - | 2 | |
| HM005 | Russian language 2 | 2 | - | 2 | HM004 |
| HM003 | Human rights& anticorruption | 2 | - | 2 | |
| 2. Elective courses (2 Credit hours) | | | | | |
| HM005 | Scientific thinking | 2 | - | 2 | |
| HM007 | Technical writing | 1 | | 1 | |
| HM009 | Scientific Research Methodology. | 2 | - | 2 | |
| HM 010 | History of Egypt | 1 | | 1 | |
| HM015 | Russian Language3 | 2 | - | 2 | HM005 |

| B: Math & Basic Science (21 credit hours) | | | | | |
|---|---|---|---|---|---|
| **Code** | **Course title(s)** | **Lectures** | **Lab/Tutorial** | **Credits** | **Pre-requisite(s)** |
| MTH101 | Mathematics 1 | 2 | -/2 | 3 | |
| MTH102 | Linear Algebra | 2 | -/2 | 3 | |
| MTH103 | Mathematics 2 | 2 | -/2 | 3 | MTH101 |
| MTH104 | Probability and Statistics 1 | 2 | -/2 | 3 | MTH101 |
| PH101 | Physics | 2 | 2/- | 3 | |
| ELC101 | Electronics | 2 | 2/- | 3 | PH101 |
| ELC201 | Logic Design | 2 | 2/- | 3 | ELC101 |
| **Math & Basic Science Elective Courses (3 Credit hours)** | | | | | |
| **Code** | **Course title(s)** | **Lectures** | **Lab/Tutorial** | **Credits** | **Pre-requisite(s)** |
| MTH201 | Mathematics 3 | 2 | -/2 | 3 | MTH103 |
| MTH202 | Probability and Statistics 2 | 2 | -/2 | 3 | MTH104 |
| MTH203 | Numerical Analysis | 2 | -/2 | 3 | MTH101 |

| C: Basic Computer Science (39 credit hours) | | | | | |
|---|---|---|---|---|---|
| **Code** | **Course title(s)** | **Lectures** | **Lab/Tutorial** | **Credits** | **Pre-requisite(s)** |
| CS101 | Introduction to Computers Science | 2 | 2/- | 3 | |
| IS101 | Fundamentals of Information Systems | 2 | 2/- | 3 | |
| CS102 | Structured programming | 2 | 2/- | 3 | CS101 |
| CS103 | Discrete Structure | 2 | 2/- | 3 | |
| IS202 | System Analysis and Design | 2 | 2/- | 3 | IS101 |
| IS201 | Database Systems | 2 | 2/- | 3 | IS101 |
| CS201 | Object oriented Programming | 2 | 2/- | 3 | CS102 |
| SW201 | Software Engineering | 2 | 2/- | 3 | CS102 |
| CS202 | Signal and System | 2 | -/2 | 3 | MTH103 |
| CS203 | Data Structures | 2 | 2/- | 3 | CS201 |
| CS204 | Operating Systems | 2 | 2/- | 3 | CS102 |
| CS205 | Algorithms | 2 | 2/- | 3 | CS102 |
| AI201 | Introduction to Artificial Intelligence | 2 | 2/- | 3 | MTH102 |

| D:Major Requirements (69 credit hours) | | | | | |
|---|---|---|---|---|---|
| 1. Compulsory Courses (45 Credit hours) | | | | | |
| **Code** | **Course title(s)** | **Lectures** | **Lab/Tutorial** | **Credits** | **Pre-requisite(s)** |
| AI301 | Machine Learning | 2 | 2/- | 3 | AI201 |
| DS301 | Data Mining& Big Data analysis | 2 | 2/- | 3 | IS201 |
| DS302 | Fundamental of Data Science | 2 | 2/- | 3 | MTH202 |
| DS303 | Exploratory Data Analysis | 2 | 2/- | 3 | MTH202 |
| DS304 | Data Visualization | 2 | 2/- | 3 | MTH202 |
| DS305 | Fundamental of Simulation | 2 | 2/- | 3 | MTH202 |
| DS306 | Advanced Database Systems | 2 | 2/- | 3 | IS201 |
| DS307 | Cloud computing | 2 | 2/- | 3 | CS204 |
| DS401 | Feature Engineering | 2 | 2/- | 3 | AI301 |
| DS402 | Advanced Machine Learning | 2 | 2/- | 3 | AI301 |
| DS403 | Applied data Science | 2 | 2/- | 3 | DS302 |
| DS404 | Social Media Analytics | 2 | 2/- | 3 | DS303 |
| DS405 | Integrated Systems and Big Data Analytics | 2 | 2/- | 3 | DS301 |
| DS406 | Graduation Project 1 | 2 | 2/- | 3 | |
| DS407 | Graduation Project 2 | 2 | 2/- | 3 | DS406 |
| 2. Elective Courses (24 Credit hours) | | | | | |
| **Code** | **Course title(s)** | **Lectures** | **Lab/Tutorial** | **Credits** | **Pre-requisite(s)** |
| CS301 | Operations Research | 2 | 2/- | 3 | MTH202 |
| CS407 | Distributed Systems | 2 | 2/- | 3 | CS204 |
| AI302 | Natural Language Processing | 2 | 2/- | 3 | AI301 |

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| AI403 | Deep Learning | 2 | 2/- | 3 | AI301 |
| IS309 | Information Retrieval and Web Search | 2 | 2/- | 3 | IS201 |
| IS310 | Business intelligence | 2 | 2/- | 3 | IS201 |
| IS411 | Business Analytics | 2 | 2/- | 3 | DS303 |
| DS308 | Data Science Tools and Techniques | 2 | 2/- | 3 | DS302 |
| DS309 | Biostatistics | 2 | 2/- | 3 | MTH202 |
| DS310 | Advanced Big Data Analysis | 2 | 2/- | 3 | DS301 |
| DS311 | Statistics for Data Science | 2 | 2/- | 3 | MTH202 |
| DS312 | Decision support for Data Science | 2 | 2/- | 3 | IS201 |
| DS313 | Applied Multivariate Analysis | 2 | 2/- | 3 | DS303 |
| DS408 | Applied Continuous Data for Regression Capstone | 2 | 2/- | 3 | DS402 |
| DS409 | Applied Discrete Data for Classification Capstone | 2 | 2/- | 3 | DS402 |
| DS410 | Text Analytics | 2 | 2/- | 3 | AI302 |
| DS411 | Applied Time Series Capstone | 2 | 2/- | 3 | DS302 |
| DS412 | Advanced Topics in Statistics | 2 | 2/- | 3 | DS309 |
| DS413 | Applied Data Science and Statistics Capstone | 2 | 2/- | 3 | DS302 |
| DS414 | Advanced Techniques of High Dimensional Data | 2 | 2/- | 3 | DS305 |
| DS415 | Web Mining | **2** | **2/-** | **3** | AI301 |
| DS416 | Multi-Objective Optimization and Decision Making | **2** | **2/-** | **3** | DS303 |
| DS417 | Selected topics in DS (1) | **2** | **2/-** | **3** | DS303 |
| DS418 | Selected topics in DS (2) | **2** | **2/-** | **3** | DS302 |

# ***Level 1***

| Code | Course title(s) | Credits |
|---|---|---|
| **Semester 1** | | |
| MTH101 | Mathematics 1 | 3 |
| MTH102 | Linear Algebra | 3 |
| PH101 | Physics | 3 |
| CS101 | Introduction to Computers Science | 3 |
| IS101 | Fundamentals of Information Systems | 3 |
| HM001 | English language 1 | 2 |
| **Total Credits** | | **17** |
| **Code** | **Course title(s)** | **Credits** |
| **Semester 2** | | |
| ELC101 | Electronics | 3 |
| MTH103 | Mathematics 2 | 3 |
| MTH104 | Probability and Statistics 1 | 3 |
| CS102 | Structured programming | 3 |
| CS103 | Discrete Structure | 3 |
| HM002 | English language 2 | 2 |
| **Total Credits** | | **17** |

# *Level 2*

| Code | Course title(s) | Credits |
|---|---|---|
| **Semester 3** | | |
| IS201 | Database Systems | 3 |
| IS202 | System Analysis and Design 1 | 3 |
| ELC201 | Logical Design | 3 |
| CS201 | Object oriented Programming | 3 |
| CS202 | Signal and System | 3 |
| HM003 | Human Rights & anticorruption | 2 |
| HM004 | Russian language 1 | 2 |
| **Total Credits** | | **19** |
| **Code** | **Course title(s)** | **Credits** |
| **Semester 4** | | |
| CS203 | Data Structure | 3 |
| CS204 | Operating Systems | 3 |
| AI201 | Introduction to Artificial Intelligence | 3 |
| CS205 | Algorithms | 3 |
| SW201 | Software Engineering | 3 |
| HM005 | Russian language 2 | 2 |
| --- | Math & Basic Science Elective Course | 3 |
| **Total Credits** | | **20** |

# ***Level 3***

| Code | Course title(s) | Credits |
|---|---|---|
| **Semester5** | | |
| DS302 | Fundamental of Data Science | 3 |
| DS303 | Exploratory Data Analysis | 3 |
| DS304 | Data Visualization | 3 |
| AI301 | Machine Learning | 3 |
| --- | Major elective 1 | 3 |
| --- | Major y elective 2 | 3 |
| **Total Credits** | | **18** |
| **Code** | **Course title(s)** | **Credits** |
| **Semester6** | | |
| DS305 | Fundamental of Simulation | 3 |
| DS301 | Data Mining& Big Data analysis | 3 |
| DS306 | Advanced Database Systems | 3 |
| DS307 | Cloud Computing | 3 |
| --- | Major elective 3 | 3 |
| --- | Major elective 4 | 3 |
| **Total Credits** | | **18** |

# ***Level 4***

| Code | Course title(s) | Credits |
|---|---|---|
| **Semester7** | | |
| DS401 | Feature Engineering | 3 |
| DS402 | Advanced Machine Learning | 3 |
| DS403 | Applied data Science | 3 |
| DS406 | Graduation Project 1 | 3 |
| ----- | Major elective 5 | 3 |
| --- | Major elective 6 | 3 |
| **Total Credits** | | **18** |
| **Code** | **Course title(s)** | **Credits** |
| **Semester 8** | | |
| DS404 | Social Media Analytics | 3 |
| DS405 | Integrated Systems and Big Data Analytics | 3 |
| DS407 | Graduation Project 2 | 3 |
| --- | Major elective 7 | 3 |
| --- | Major elective 8 | 3 |
| -- | University elective 1 | 2 |
| **Total Credits** | | **17** |

# Courses Descriptions

| University Requirements Courses | | |
|---|---|---|
| HM001 | English language 1 | 2 Cr. Hrs |
| Introduction – characteristics of technical English language – revision of English grammar – some styles of writing – characteristics of effective sentences – common faults in writing of sentences in English language – construction of paragraphs: main idea – methods of presentation of main idea – types of paragraphs. In addition, reading activities and general vocabulary. | | |
| HM002 | English language 2 | 2 Cr. Hrs |
| Exercises related to scientific topics – Development of student's knowledge in language and ability for reading and understanding – Developing of understanding and translation abilities of the student – Developing of listening and speech abilities of the student – exercises on writing technical topics – readings in scientific books – methods of search – technical writing. | | |
| HM003 | Human rights& anticorruption | 2 Cr. Hrs |
| This course contains an overview of anti-corruption. It aims to raise awareness and deal with corruption as a matter of public policy. It also includes an introduction to basic human rights under international and national law. This course includes interactive parts that were produced electronically by the National Center for E-Learning of the Supreme Council of Universities. | | |
| HM004 | Russian language 1 | 2 Cr. Hrs |
| Introduction to phonetics. Idea of stress. System of consonants and vowels. Differentiation between sound and letter. Phonetic rules of pronunciation of vowels "o" and "e" while unstressed. Pronunciation of voiced consonants as voiceless at the end of the word. | | |
| HM005 | Russian language 2 | 2 Cr. Hrs |
| This courses covers Development of conversational skills on such topics as "meeting people", "introduction of oneself", "family". Grammatical constructions with such expressions as "How many?", "What time is it now?", "How old are you?", "How much?" Present tense. Genitive case. Personal pronouns. | | |
| **University Elective courses** | | |
| HM006 | Scientific thinking | 2 Cr. Hrs |
| This course is required for all students. It emphasizes the unifying aspects of the scientific approach to the study of nature and human behavior. About one-third of the course is devoted to a discussion of the nature of scientific inquiry and investigation. The course focuses on the process of fact identification and concept formation and testing. In the remainder of the course students are exposed to applications of the approach in various disciplines. The course sets some of the major concepts and theories of science into a broad historical, philosophical, and cultural context and traces the development of these theories and concepts to their present status. This serves the double purpose of acquainting the students with the appropriate setting in which a given | | |

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| idea gains relevance and exposing them to the evolution of current methods of investigation. | | |
| HM007 | Technical writing | 1 Cr. Hrs |
| This course contains an introduction to writing numbers, signs, abbreviations and equations. The course includes teaching the student the basics of writing paragraphs (Paragraph), how to start writing, writing the summary and summary, and organizing writing for the topic, as well as reviewing and editing texts. This course provides the student with instruction in various forms of writing, such as correspondence in the field of specialization, proposals, reports, and examples References and table. As well as how to make a final writing review | | |
| HM008 | Scientific Research Methodology. | 2 Cr. Hrs |
| The course is designed to provide students a solid foundation for designing and conducting their own research and to enable them to become knowledgeable individuals of prior academic research results. The course provides an introduction to scientific research methodology and techniques. The course contains the main issues related to research design; quantitative and qualitative research approaches and related methods; selected number of research techniques in social sciences. The course also provides training to students about how to prepare a research proposal prior to conducting their graduation project (dissertation). | | |
| HM009 | History of Egypt | 1 Cr. Hrs |
| This course examines the history of ancient Egypt from the rise of the Egyptian state to the end of the 18th Dynasty in the New Kingdom, a period of more than 1500 years. This span encompasses three separate “golden ages” of Egyptian history and culture, from the pyramid builders of the Old Kingdom through the Middle Kingdom renaissance to the expansionist era of the early New Kingdom. By looking at both political and social history this course will allow students to examine the differences as well as the similarities between these periods and come to a better understanding of the factors, internal and external, which influenced Egyptian historical development. | | |
| HM010 | Russian Language3 | 2 Cr. Hrs |
| Development of listening, conversational, reading and writing skills. System of tenses in Russian grammar. Accusative case constructions. Instrumental case constructions. Personal and possessive pronouns. Main purpose – development of conversational skills on the basis of grammar material. Such topics as “university”, “shop”, “library”, “hospital”. Information about countries. Such topics as “Moscow”, “Cairo” | | |

| Math & Basic Science Courses | | |
|---|---|---|
| MTH101 | Mathematics 1 | 3 Cr. Hrs |
| The course contains, Limits and continuity, Differentiation, Taylor's series, parametric curves, Partial Differentiation, trigonometric functions, Applications of differentiation, Integration, Techniques of integration, improper integral, applications of integration, and vectors in the plane. | | |
| MTH102 | Linear Algebra | 3 Cr. Hrs |
| This course concentrates on the mathematical theory and methods of linear algebra. This course contains: Indeterminate forms, Infinite series, vectors, curves and surfaces in space, Binomial theorem, matrices and operations, homogenizes and non-homogenous linear equations, Determinants, vector spaces and subspaces, eigenvalues and eigenvectors, matrix diagonalization, partitioned matrices, quadratic forms, and applications of linear algebra. | | |
| MTH103 | Mathematics 2 | 3 Cr. Hrs |
| Basic definition of differential equations- First order differential equations –second and higher order differential equations- Laplace Transformation- Solving differential equations using Laplace Transform- Solving differential equations using series-systems of differential equations- Fourier series- Partial differential equations- Maxima and Minima of functions of two variables- Double Integral and Triple Integral and its applications. | | |
| MTH104 | Probability and Statistics 1 | 3 Cr. Hrs |
| The course contains descriptive statistics, Visual representations of data, Probability theory and statistical inference, random variables, and probability distributions. Estimation and confidence intervals, Correlation, and regression. | | |
| PH101 | Physics | 3 Cr. Hrs |
| The course contains the electric field, Gauss's law, electric potential, capacitance, DC circuits, magnetic fields, Faraday's and Ampere's laws, time-varying fields, Maxwell's equations in integral form, and alternating currents, which earns the student cognitive and technical skills in physics. | | |
| ELC101 | Electronics | 3 Cr. Hrs |
| Introduction, Insulators, Conductors, Semiconductors, Resistors, Capacitors, Coils, Diodes, Application of diodes, Rectifiers, Power supplies, LEDs, Voltage regulators, Transistors, Amplifiers, Timers, Applications. | | |
| ELC201 | Logic Design | 3 Cr. Hrs |

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| This course contains all topics related to logical design, such as a Review on number systems, Coded number systems, logical functions, and their minimization, NAND and NOR gates, Mutual circuit design, adders, subtractors, decoders, encoders, multiplexer/demultiplexer, memory elements: latch and SR-FF, JK-FF, D-FF, T-FF, counters, programmable logic design (PLD) and (FPGA). | | |
| **Math & Basic Science Elective Courses** | | |
| MTH201 | Mathematics 3 | 3 Cr. Hrs |
| This course contains basic definitions of the scalar product of two vectors- the cross product of two vectors- Triple scalar (mixed) product of three scalars- scalar functions and vector functions- vector field- derivative differential operator- the divergence of a vector- curl of a vector- line integrals - surface integral- Gauss divergence theorem - Stokes theorem- Complex functions- Analytic functions- Complex integral. | | |
| MTH202 | Probability and Statistics 2 | 3 Cr. Hrs |
| Types of statistics, Population versus sample, basic terms - numerical descriptive measures for populations and samples, for ungrouped and grouped data: Measures of Central Tendency, Measures of Dispersion, And Measures of Position –Some special discrete distribution( Binomial-Geometric-Negative-Hyper geometric-Poisson, etc.) and continuous distributions (Uniform-Exponential-Weibull-Gamma-Beta-Normal-log normal-Cauchy) , sampling distribution for the sample means – sample proportions and sample variations-center limit theorem – hypothesis testing. | | |
| MTH203 | Numerical Analysis | 3 Cr. Hrs |
| This course contains Numerical Computing and Computers - Solving of Linear systems- Solving Non-linear Equations-Interpolation and curve fitting- approximation of Functions-Finite Difference Numerical Differentiation and Numerical Integration- Numerical solution of ordinary differential equations. | | |

| C: Basic Computer Science (39 credit hours) | | |
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| CS101 | Introduction to Computers Science | 3 Cr. Hrs |
| This course is designed to introduce various types of computers, Computer hardware and software components, different between hardware and software, numbering systems which include binary, decimal, octal, and hexadecimal system. In addition, the course will cover introduction to computer networks, and the main components and protocols used to access the internet. | | |
| IS101 | Fundamentals of Information Systems | 3 Cr. Hrs |
| This course includes the following topics: the components of information systems, the different types of information systems, the Information Technology (IT) and its impact on information systems, the systems development life cycle (SDLC), and how IS effect on competitive advantages in business. | | |
| CS102 | Structured programming | 3 Cr. Hrs |
| Fundamental programming constructs: Syntax and semantics of a higher-level language; variables, types, expressions, and assignment − Simple I/O − Conditional and iterative control structures − Functions and parameter passing − Structured decomposition − Algorithms and problem-solving: Problem-solving strategies; the role of algorithms in the problem-solving process; implementation strategies for algorithms; debugging strategies. | | |
| CS103 | Discrete Structure | 3 Cr. Hr |
| Introduction to logic and proofs − Fundamental structures: Functions; relations; sets; cardinality and countability − Boolean algebra − Propositional logic: Logical connectives; truth tables; normal forms; validity − Elementary number theory: Factorability; properties of primes; greatest common divisors and least common multiples; Euclid's algorithm; modular arithmetic; the Chinese Remainder Theorem − Basics of counting: Counting arguments; pigeonhole principle; permutations and combinations; binomial coefficients − Predicate logic: Universal and existential quantification; modus ponens and modus tollens; limitations of predicate logic − Recurrence relations: Basic formulae; elementary solution techniques − Graphs and trees: Fundamental definitions; simple algorithms; traversal strategies; proof techniques; spanning trees; applications. | | |
| IS202 | System Analysis and Design | 3 Cr. Hrs |
| The aim of this course is to cover the topics related to the upper phases of the Information Systems Development Life Cycle, which are the following: Planning of IS, the detailed analysis of IS, and the Conceptual design of IS. Methodologies, techniques, activities, tasks, deliverables and practical experiences related to the execution of these phases will be covered in the course. | | |
| IS201 | Database Systems | 3 Cr. Hrs |
| This course is the main course in information system. In this course students learn the concepts of database and the difference between file database and relational database. This course also covers the following topics: Phases of database design, Conceptual database design, Classification, specialization, and aggregation abstraction, The Entity-Relationship model, Extended Entity-Relationship model, View design in conceptual schema, Conceptual schema integration, Transforming conceptual schema to relations, Logical database design, Characteristics of good relation schema, Anomalies in relational schema, Function | | |

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| dependencies, and Normalization. The student will learn SQL basic and the advanced concepts of Structured Query Language (SQL). | | |
| CS201 | Object oriented Programming | 3 Cr. Hrs |
| This course contains object-oriented programming principles and techniques. Topics include object-oriented programming concepts, such as classes, objects, methods, interfaces, packages, inheritance, encapsulation, polymorphism, and information hiding. The student will apply techniques using a modern object oriented implementation language. | | |
| SW201 | Software Engineering | 3 Cr. Hrs |
| This course provides an overview of software engineering. This course contains the following topics: Software development life cycle, Project teams, documentation, and group dynamics. Software cost estimation, Requirements engineering and design, Data modeling, object modeling, and object-oriented analysis, Object-oriented programming and design, Software testing, quality assurance, and Software maintenance. | | |
| CS202 | Signal and System | 3 Cr. Hrs |
| Digital Signal Processing (DSP) is concerned with representing signals in digital form and processing these signals and the information they carry. Topics includes: Introduction to continuous time and discrete time signals and systems, linear time invariant systems, Fourier series representation for continuous and discrete time signals, Fourier transform for continuous and discrete time signals, Sampling theorem, Laplace transform, Z-Transform, transfer function, Digital signal convolution, Digital filter design: Finite impulse response, Infinite impulse response. Adaptive digital filters: Concepts, Algorithms, Applications. | | |
| CS203 | Data Structures | 3 Cr. Hrs |
| This course contains topics that include: built-in data structures. Stacks, queues, linked lists, and tree structures. Sorting algorithms, searching algorithms, and hashing. Abstract data types, this course helps students to analyze, evaluate and deal with data. | | |
| CS204 | Operating Systems | 3 Cr. Hrs |
| This course examines types of operating systems. Operating Systems structures: system components and services, also covers the classical internal algorithms and structures of operating systems, including CPU scheduling, memory management, and device management. Considers the unifying concept of the operating system as a collection of cooperating sequential processes. Covers topics including file systems, virtual memory, disk request scheduling, concurrent processes, deadlocks, security, and integrity | | |
| CS205 | Algorithms | 3 Cr. Hrs |
| This course is concerned with how to think to solve the problems and how to analysis and design algorithms. In this course the students learn different techniques for designing algorithms, advanced sorting and searching methods, Graph algorithms, Notion of an algorithm, Space and time complexities of an algorithm. In addition, they will aware with Fundamental design paradigms: Divide and conquer, Branch and bound, Backtracking dynamic programming greedy methods and Simulation. binary search trees; representations of graphs; depth- and breadth-first traversals – Recursion and divide-and-conquer strategies – Basic algorithmic strategies: Brute-force algorithms; greedy algorithms. | | |
| AI201 | Introduction to Artificial Intelligence | 3 Cr. Hrs |

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| The aim of the module is to introduce students to the principal theory and techniques of Artificial Intelligence – AI and build an awareness of its applications and current research trends. The module also aims to develop a broad appreciation of the nature of the problems associated with AI and introduce common techniques that are used to address them. |

| Major Requirements Courses | | |
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| DS302 | Fundamental of Data Science | 3 Cr. Hrs |
| This course introduces the fundamentals of the data science domain. Fundamentals of Python programming in the context of Data Science with a focus on relevant packages. Coverage of techniques for database handling, data manipulation, visualization, and summarization. Study of probability basics and further statistics required for assessing sampling techniques, designing estimators, hypothesis testing and fitness tests. Focus on simulation basics, analytics, data interpretation and basic machine learning models such as regression and kNN classifiers. | | |
| DS303 | Exploratory Data Analysis | 3 Cr. Hrs |
| This course introduces a set of methods used to analyze, summarize, and examine the data before building a model. Data Scientists and Analysts try to find different patterns, relations, and anomalies in the data using some statistical graphs and other visualization techniques. Topics as Get maximum insights from a data set, Uncover underlying structure, Extract important variables from the dataset, Detect outliers and anomalies (if any), Test underlying assumptions, and Determine the optimal factor settings. Learning the major tools used for preparing the data for analysis with Python using NumPy and Pandas Libraries. | | |
| DS304 | Data Visualization | 3 Cr. Hrs |
| This course discusses visual representation methods and techniques that increase the understanding of complex data. Good visualizations not only present a visual interpretation of data, but do so by improving comprehension, communication, and decision making. In this course, you will learn how the human visual system processes and perceives images, good design practices for visualization, tools for visualization of data from a variety of fields, and programming of interactive visualizations using 3D. Learning the major tools used for preparing the data for analysis with Python using Matplotlib and Seaborn Libraries. | | |
| DS305 | Fundamental of Simulation | 3 Cr. Hrs |
| This course is an introduction to fundamental tools in designing, conducting, and interpreting simulation experiments. Topics covered include Random number generation; Continuous, discrete, and rare event simulations. Variance reduction techniques, Bootstrap; Simulation; Markov Chain Monte Carlo. The course includes an applied project (a thorough application of simulation to real-world problems such as finance, statistics) using Python programming language). | | |
| AI301 | Machine Learning | 3 Cr. Hrs |
| This course provides a broad introduction to machine learning and statistical pattern recognition. Topics include: supervised learning (generative/discriminative learning, parametric/non-parametric learning, neural networks, support vector machines); unsupervised learning (clustering, dimensionality reduction, kernel methods); learning theory (bias/variance tradeoffs, practical | | |

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| advice); reinforcement learning and adaptive control. The course will also discuss recent applications of machine learning | | |
| DS306 | Advanced Database Systems | 3 Cr. Hrs |
| This course provides the advanced relational database theory: functional dependencies, multivalued dependencies, join dependencies, inclusion dependencies. System catalog implementation, query optimization techniques, transaction processing, concurrency control, database security, backup, and recovery strategies. Advanced data modeling (e.g., object-oriented databases), distributed and client-server architectures, and further exposure to social and ethical issues in databases. | | |
| DS307 | Cloud computing | 3 Cr. Hrs |
| The course presents a top-down view of cloud computing, from applications and administration to programming and infrastructure. Its main focus is on parallel programming techniques for cloud computing and large scale distributed systems which form the cloud infrastructure. The topics include: overview of cloud computing, cloud systems, parallel processing in the cloud, distributed storage systems, virtualization, security in the cloud, and multicore operating systems. Students will study state-of-the-art solutions for cloud computing developed by Google, Amazon, Microsoft, Yahoo, VMWare, etc. Students will also apply what they learn in one programming assignment and one project executed over Amazon Web Services. | | |
| DS312 | Decision support for Data Science | 3 Cr. Hrs |
| This course seeks research that promotes technical, theoretical, design science, pedagogical, and behavioral research as well as emerging applications in analytics and big data. Topics include data analytics & visualization from varied data sources such as sensors or IoT data, text, multimedia, clickstreams, user-generated content involving issues dealing with curation; management and infrastructure for (big) data; standards, semantics, privacy, security and legal issues in big data, analytics and KM (knowledge management); intelligence and scientific discovery using big data; analytics applications in various domains such as smart cities, smart grids, financial fraud detection, digital learning, healthcare, criminal justice, energy, environmental and scientific domains, sustainability and the like; business process management applications such as process discovery, performance analysis, process conformance and mining using analytics and KM, cost-sensitive, value-oriented, and data-driven decision analysis and optimization. | | |
| DS301 | Data Mining& Big Data analysis | 3 Cr. Hrs |
| This course introduces the fundamental concepts of Data Mining, and Big Data motivation and applications, text and web mining, and survey of techniques and models. Topics include data pre-processing, frequent pattern mining, clustering, classification, and case studies using data sets taken from real-life applications, and use of data mining software. | | |
| DS401 | Feature Engineering | 3 Cr. Hrs |
| This course provides the main step for feature engineering. Topics as Learning multiple techniques for missing data imputation, Transform categorical variables into numbers while capturing meaningful information, Learn how to deal with infrequent, rare, and unseen categories, Learn how to work with skewed variables, Convert numerical variables into discrete ones, Remove outliers from your variables, Extract useful features from dates and time variables, Learn techniques used in organizations worldwide and in data competitions, and Increase your repertoire of techniques to pre-process data and build more powerful machine learning models. | | |
| DS402 | Advanced Machine Learning | 3 Cr. Hrs |

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| This course includes the Perceptron and multilayer neural networks, Gradient descent, Back-propagation, Convolutional Neural Network (CNN), pooling layers, CNN applications, Recurrent Neural Networks (RNN), vanishing gradient problem, case studies: object detection, and stock price estimation, GANs. | | |
| DS403 | Applied data Science | 3 Cr. Hrs |
| This course aims at introducing students to novel topics and applications in data science that need to be identified in a responsive manner as technology and needs evolve and develop including: blockchain, health informatics, intelligent transportation, image recognition, search engines, natural language processing, etc. | | |
| DS404 | Social Media Analytics | 3 Cr. Hrs |
| This course represents an introduction to analyzing social media data with an emphasis on the implications for communication strategy, market research, and technology history. How do we know whether a feature on the new iPhone is working? What type of consumers are most likely to buy a given product? How does the perception of a company change after a crisis? Social media streams can provide valuable insights into users, brands, audiences, and industries. The course is composed of two primary parts: (1) an overview of the approaches and challenges associated with doing social media research, and (2) an involved project in which you will apply the knowledge gained to a self-determined case study on a social platform. | | |
| DS405 | Integrated Systems and Big Data Analytics | 3 Cr. Hrs |
| This course examines information management in the context of massive sets of data, provides students proficiency with a variety of data analysis tools, and exposes learners to varied data platforms as well as skills and concepts related to data mining and statistical analysis within a business context. The focus of the course is on the wide range of technologies available today as well as the methodologies of data analytics for solving Big Data problems. This course explores the role of business analytics in supporting decision-making and setting strategies at the highest levels. Students will be exposed to the analytical skills needed to turn structured and unstructured big data into a strategic resource. Students will work on application areas that create or use big data and will be expected to work on relevant projects mainly focusing on the 5 key Big Data use cases: Big Data exploration enhanced 360 views of the customer, security/intelligence, extension, operations analysis, and data warehouse augmentation. | | |
| DS406 | Graduation Project 1 | 3 Cr. Hrs |
| In this course, students will work on a research problem in Data Science, bringing to bear the tools and techniques that they have learned to investigate a novel aspect of Data Science. This is an independent project, supervised by an expert from the relevant area, and culminates in writing a dissertation in the form of a research paper, describing your research and its results. Research topics can be selected from across the breadth of Data Science, and the project may include theoretical analysis, as well as practical software implementation. | | |
| DS407 | Graduation Project 2 | 3 Cr. Hrs |
| In this course, students will work on a research problem in Data Science, bringing to bear the tools and techniques that they have learned to investigate a novel aspect of Data Science. This is an independent project, supervised by an expert from the relevant area, and culminates in writing a dissertation in the form of a research paper, describing your research and its results. Research topics can be selected from across the breadth of Data Science, and the project may include theoretical analysis, as well as practical software implementation. | | |

| Major Elective Courses | | |
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| CS301 | Operations Research | 3 Cr. Hrs |
| This course helps students for solving problems in different environments that needs decisions. The course cover topics that include: linear programming, Transportation, Assignment, Critical Path Method (CPM) and Project Evaluation and Review technique (PERT). In applications computer packages will be used for business managers and decision making. | | |
| DS310 | Advanced Big Data Analysis | 3 Cr. Hrs |
| The course introduces Big Data problems and associated frameworks and technologies. First, the course motivates the topic using real-world big data problems. Second, it sheds light on handling big data, from data collection, to monitoring, storage, analysis and reporting. The course also includes programming models used for scalable big data analysis. It also introduces one of the most common Big Data frameworks, namely Hadoop, in addition to the Map-Reduce Programming Model. Finally, it solves sample case studies using the covered Big Data analytics tools. | | |
| DS309 | Biostatistics | 3 Cr. Hrs |
| This course is an introduction to important topics in biological, medical, health, and environmental statistical concepts and reasoning. Topics include Introduction to Biological data Processing and Analysis, Hypothesis testing, nonparametric tests, Logistic regression, Poisson regression, Statistical methodologies in the analysis of survival data (Kaplan-Meier estimator, Cox's proportional hazards models, time-dependent covariates, multiple failure outcomes). Typical biomedical applications, including clinical trials. The course includes an applied project using the computer programming language Python. | | |
| IS411 | Business Analytics | 3 Cr. Hrs |
| This course aims to give an exploration and investigation of past business performance based on data and statistical methods. It studies statistical methods that can be applied to a specific project, process or product to evaluate the business. This course provides an extensive use of statistical analysis, explanatory, predictive modeling, and fact-based management which help in decision making. Analytics may be used as input for human decisions or may drive fully automated decisions | | |
| DS308 | Data Science Tools and Techniques | 3 Cr. Hrs |
| This course explores the data science lifecycle, including question formulation, data collection and cleaning, exploratory data analysis and visualization, statistical inference and prediction, and decision-making. It focuses on quantitative critical thinking and the key principles and techniques that are needed. These include languages for transforming, querying and analyzing data; algorithms for machine learning methods including regression, classification and clustering; principles behind creating informative data visualizations; statistical concepts of measurement error and prediction; and techniques for scalable data processing. | | |
| DS313 | Applied Multivariate Analysis | 3 Cr. Hrs |

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| Techniques of multivariate statistical analysis illustrated by examples from various fields. Topics include Multivariate normal distribution. Sample geometry and multivariate distances. Inference about a mean vector. Comparison of several multivariate means, variances, and covariances. Detection of multivariate outliers. Principle components. Multidimensional scaling. Factor analysis. Canonical correlation. Discriminant analysis. and Clustering. The course includes an applied project (a thorough analysis of real-life data sets using computer-packaged programs). | | |
| DS408 | Applied Continuous Data for Regression Capstone | 3 Cr. Hrs |
| This Course includes an applied project (a thorough analysis of real-life data using computer packaged programs). Topics as Standard least squares method and application to problems arising from social, biological, and engineering sciences. Deviation from assumption of multicollinearity. Variable selection methods. Analysis of variance, Generalized linear models including logistic regression models. | | |
| DS409 | Applied Discrete Data for Classification Capstone | 3 Cr. Hrs |
| The analysis of discretely measured responses such as counts, proportions, nominal variables, ordinal variables, discrete interval variables with few values, continuous variables grouped into a small number of categories, etc. Topics include: Detection of outliers in categorical data; Inference and distributions for discrete data; Inference for contingency tables; Generalized linear models for count data; Models for binary and multinomial response (logistic regression, Poisson regression, Loglinear models); Classification and regression trees; Cluster analysis for categorical data. The course includes an applied project using computer programming language Python. | | |
| IS310 | Business intelligence | 3 Cr. Hrs |
| In this course, you will gain the knowledge and skills for using data warehouses for business intelligence purposes and for working as a business intelligence developer. You'll have the opportunity to work with large data sets in a data warehouse environment and will learn the use of MicroStrategy's Online Analytical Processing (OLAP) and Visualization capabilities to create visualizations and dashboards.<br><br>The course gives an overview of how business intelligence technologies can support decision making across any number of business sectors. | | |
| IS309 | Information Retrieval and Web Search | 3 Cr. Hrs |
| This course develops the competencies and skills for planning and controlling projects and Information retrieval is the process through which a computer system can respond to a user's query for text-based information on a specific topic. IR was one of the first and remains one of the most important problems in the domain of natural language processing (NLP). Web search is the application of information retrieval techniques to the largest corpus of text anywhere — the web — and it is the context where many people interact with IR systems most frequently.<br><br>In this course, we will cover basic and advanced techniques for building text-based information systems, including the following topics:<br><br>• Efficient text indexing<br>• Boolean and vector-space retrieval models<br>• Evaluation and interface issues<br>• IR techniques for the web, including crawling, link-based algorithms, and metadata usage | | |

- Document clustering and classification
- Traditional and machine learning-based ranking approaches

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| AI302 | Natural Language Processing | 3 Cr. Hrs |

This course contains an overview of NLP with a major focus on probability and machine learning techniques. The machine learning-based methodology applied to language data with assessment of these experiences is essential. If the course includes an overview of typical NLP applications, such as information extraction, machine translation, question-answer systems, and more focus on one of these applications. In addition, a typical NLP system, such as marking, analysis, and identification of the named entity, also extracts the relationship.

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| DS410 | Text Analytics | 3 Cr. Hrs |

This course gives the fundamental concepts of text analytics and performs text analytics on different applications. Investigates linguistic, statistical, and machine learning techniques for modeling the information in textual sources. Includes information retrieval, natural language processing, text classification, and sentiment analysis and the software systems for performing these analyses.

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| DS415 | Web Mining | 3 Cr. Hrs |

Web mining aims to discover useful information and knowledge from the Web hyperlink structure, page contents, and usage logs. It has direct applications in e-commerce, Web analytics, information retrieval/filtering, personalization, and recommender systems. Employees knowledgeable about Web mining techniques and their applications are highly sought by major Web companies such as Google, Amazon, Yahoo, MSN, and others who need to understand user behavior and utilize discovered patterns from terabytes of user profile data to design more intelligent applications. The primary focus of this course is on Web usage mining and its applications to business intelligence and biomedical domains. We learn techniques from machine learning, data mining, text mining, and databases to extract useful knowledge from the Web and other unstructured/semi-structured, hypertextual, distributed information repositories. This data could be used for site management, automatic personalization, recommendation, and user profiling. Topics covered include crawling, indexing, ranking, and filtering algorithms using text and link analysis, applications to search, classification, tracking, monitoring, and Web intelligence. Programming assignments give hands-on experience. A group project highlights class topics.

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| CS407 | Distributed Systems | 3 Cr. Hrs |

The course contains the key characteristics of Distributed Systems (DSs), Client/Server Computing; Why client-server computing, Down-sizing the main frame, Distributed access to SQL data. Inter-process communication, UNIX case study. Remote Procedure Calling; Distributed data and distributed processing, Applications: Distributed File Systems, Binding and Name Servers, E-mail, GroupWare. Shared data and Transactions; Distributed DB, Transaction processing, Conversation between client and server, and Concurrency control.

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| DS411 | Applied Time Series Capstone | 3 Cr. Hrs |

This course deals with the problems of modeling and forecasting time series data. Computer program packages are used as an aid for obtaining solutions. Topics include serial correlation, seasonal adjustments, exponential smoothing and extrapolation, state-space models, moving average, autoregressive, ARMA and ARIMA models, and nonlinear time series, including ARCH models and chaos. Emphasis on model building, diagnostic checking, and model selection.

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| DS412 | Advanced Topics in Statistics | 3 Cr. Hrs |
| This course offers an insight into cutting-edge statistical learning techniques that are at the forefront of current research and application. You will have the opportunity to explore a range of topics including time series modeling and forecasting, decision trees, random forests, support vector machines, neural networks, and Bayesian computation. The choice of topics in any year may change to ensure that the content of the module reflects the rapid change in this exciting area. | | |
| DS413 | Applied Data Science and Statistics Capstone | 3 Cr. Hrs |
| In this course, students will work on a research problem in the application of Data Science and Statistics. They will apply your understanding of the underlying concepts of Data Science and Statistics together with the methods and tools that you have learned to a problem in an applied field. The project will require an understanding of the setting, a critical review of possible approaches, a choice of appropriate methodology, an extended piece of data analysis, and a clear and concise write-up of the background, data, methodology, results, and conclusions. This is an independent project, supervised by an expert from the relevant area, and culminates in writing a dissertation, describing your research and its results. Research topics can be selected from across the breadth of the application of Data Science and Statistics. | | |
| DS414 | Advanced Techniques of High Dimensional Data | 3 Cr. Hrs |
| In this course we will learn the selected advanced topics from spatial & multimedia databases: multidimensional data management concepts, theories, and technologies, focusing on data access methods and similarity query processing for spatial, multimedia, and Web-based databases, with particular emphasis on video indexing and search. | | |
| DS311 | Statistics for Data Science | 3 Cr. Hrs |
| This course introduce the basic principles of statistical methods and procedures used for data analysis, data gathering, summarizing data using descriptive statistics, displaying and visualizing data, examining relationships between variables, probability distributions, expected values, hypothesis testing, introduction to ANOVA (analysis of variance), regression and correlation analysis.<br>• Calculate and apply measures of central tendency and measures of dispersion to grouped and ungrouped data.<br>• Summarize, present, and visualize data in a way that is clear, concise, and provides a practical insight for non-statisticians needing the results.<br>• Identify appropriate hypothesis tests to use for common data sets.<br>• Conduct hypothesis tests, correlation tests, and regression analysis.<br>• Demonstrate proficiency in statistical analysis using Python and Jupyter Notebooks. | | |
| AI403 | Deep Learning | 3 Cr. Hrs |
| In this course, Students become familiar with the fundamental concepts and terminology used in deep learning, and understand why deep learning techniques are so powerful today. Students build simple neural networks yourself and generate predictions with them. Then they will learn how to optimize the predictions generated by your neural networks. They will do this using a method called backward propagation, which is one of the most important techniques in deep learning. Understanding how it works will give you a strong foundation to build from in the second half of the course. Finally, students use the keras library to build deep learning models for both regression as well as classification! Students also learn about the Specify-Compile-Fit workflow that you can | | |

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| use to make predictions and by the end of this chapter, students will have all the tools necessary to build deep neural networks. | | |
| DS416 | Multi-Objective Optimization and Decision Making | 3 Cr. Hrs |
| Throughout industry and science, optimization tasks require the trading-off of multiple quality criteria which are in competition with one another. This multi-objective optimization task often requires the search and return of a set of solutions rather than a single design. This module spans specialized 'expensive' optimization approaches where the cost function can only be queries a few hundred times at most, through to those designed for real-time optimization of multi-objective optimization problems which change over time and robust optimization. It also covers multi-criterion decision making – which is concerned with how a final design is selected from a trade-off set. | | |
| DS417 | Selected topics in DS (1) | 3 Cr. Hrs |
| Topics are chosen according to the special interests of faculty and major students. May be repeated for credit more than once if content changes. | | |
| DS418 | Selected topics in DS (2) | 3 Cr. Hrs |
| Topics are chosen according to the special interests of faculty and major students. May be repeated for credit more than once if content changes. | | |

# Seventh: Basic concepts and terms

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| 1 | Institution vision | The vision of the institution is a brief description of what the institution aspires to achieve in the future, and it aims at the society of the institution. |
| 2 | Institution message | The mission of the institution is a paragraph that expresses what the organization is, why it exists, and defines the scope and how of its operations and what it provides to society. |
| 3 | Graduate specifications | A set of specifications determined by the educational institution by adopting academic standards and achieved by the educational program in the graduate according to the knowledge and skills it provides to students in light of the professional needs, requirements and expectations of society and the labor market. |
| 4 | National Academic Standards of Reference (NARS) | Academic standards for the various educational programs prepared by the commission with the assistance of specialized experts and representatives of the various sectors of the beneficiaries. These standards represent the minimum required to be achieved for accreditation. |
| 5 | Educational programs | It is implemented in higher education institutions, and aims to obtain a specific academic degree (for example, a Bachelor's/Bachelor/Master/PhD), and it includes curricula, courses, and activities that provide the student with the knowledge, skills, and values necessary to achieve planned educational goals, and in a specific academic specialization. Usually a higher education institution offers one or more educational programs. |
| 6 | Credit hour | A credit hour is an academic unit of measurement to determine the course weight among other courses, and it is equivalent to an hour-long theoretical lecture, a training session, or a practical lesson of at least two hours. |